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Research paper

Numerical modeling of gas flow and solid deformation in unconventional anisotropic shale

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ABSTRACT

Since coupled flow and solid deformation were introduced into shale gas reservoir simulations, the geomechanical effects on gas production have become a frequently studied topic. While numerous studies suggest that geomechanical effects tend to reduce cumulative gas production, an apparent contradiction arises when considering that elastoplastic deformation could cause additional pore compaction during production, compared to a rigid porous matrix. Specifically, assuming identical initial and final gas pressures, an isothermal process, and ideal gas behavior, such additional compaction should theoretically enhance ultimate gas production. Furthermore, widely used shale gas reservoir simulation approaches have limitations in simultaneously modeling mechanical anisotropy and elastoplastic deformation. To address these contradictions and challenges, this work develops an in-house simulator based on a rigorous mixed $\boldsymbol{u}/p$ formulation coupled with a one-dimensional finite element discretization of the fracture gas flow equation, which has been validated rigorously and demonstrates satisfactory performance. A main improvement lies in the incorporation of a compressible gas in the pores rather than a slightly compressible fluid such as water, together with the associated numerical ramifications. Numerical simulations performed on intact shale samples and field-scale fractured shale reservoirs not only address the contradictions mentioned above but also uncover an unreported trend in the plastic strain distribution during gas production associated with different bedding plane orientations. The impact of the fluid type is also highlighted.

1. Introduction

Shale, as an essential geomaterial with wide-ranging applications in both energy production and civil infrastructure, has garnered significant attention in recent years (Borja et al., 2020). Understanding the mechanical behavior of shale is crucial for optimizing extraction techniques in the energy sector and ensuring the stability of structures built on shale formations (Chen et al., 2023; Liu and Borja, 2022). Numerical simulation has proven to be a valuable tool in unraveling the intricate mechanics of shale in various scenarios, such as fault slip and earthquake triggering (Jha and Juanes, 2014; Zhao and Jha, 2021), hydraulic fracturing (Yuan et al., 2025; Zeng et al., 2020), shale gas production (Liu et al., 2020; Lu et al., 2021; Yan et al., 2018a; Yang et al., 2023), wellbore stability (Ma et al., 2025b), strip load on shale foundation (Zhao and Borja, 2021), tunnel stability in rock masses (Ip et al., 2021; Xu et al., 2021; Zhao et al., 2025), etc. These processes exhibit multiscale and multiphysics nature, driving the development of increasingly sophisticated numerical models in recent decades.

For simulating flow and deformation in fractured shale reservoir, most of the literature focused on the multi-scale nature of the porous structure (Micheal et al., 2022), in which the hydraulic fracture or macro-fracture is modeled by the discrete fracture model (DFM) (Garipov et al., 2016) or embedded discrete fracture model (EDFM) (Yan et al., 2018a), and the hierarchical porous matrix is described using equivalent single porosity model (Liu et al., 2019; Wang et al., 2022a), double porosity model (Liu and Borja, 2022), or multiple porosity model (also known as the Multiple Interacting Continua, or MINC for short), which considers the micro-fracture, micro-pores, inorganic matter/clay pores, and organic matter/kerogen pores (Mehrabian and Abousleiman, 2018). Despite these exciting achievements, two aspects of limitations remain, one on the intrinsic anisotropy, and the other on the fluid type in the finite element discretization, which are detailed in the following two paragraphs.

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Anisotropy is a ubiquitous property of shale (Yin et al., 2021; Zhao et al., 2025), which has been successfully demonstrated through laboratory experiments (Villamor Lora et al., 2016). The anisotropy could be embedded into both the permeability tensor (Clavaud et al., 2008; Ma et al., 2025a) and the elastic or elastoplastic tensor (Lonardelli et al., 2007; Semnani et al., 2016; Zhao et al., 2018). However, the anisotropy is still not frequently considered in the unconventional reservoir simulation until now. We think there are three potential causes, as listed below.

1. A majority of reservoir simulators utilize the Two-Point Flux Approximation technique, commonly denoted as TPFA for short. However, this approach handles permeability anisotropy poorly unless the structured grid is aligned with the permeability tensor's principal directions (Karimi-Fard and Durlofsky, 2016). Although alternatives such as Multi-Point Flux Approximation (MPFA) and Mimetic Finite Difference (MFD) methods exist (Sandve et al., 2012; Yan et al., 2021), their computational costs are typically much higher than those of TPFA.
2. Most reservoir simulators would handle solid deformation through a simple rock compressibility term (Gutierrez et al., 2001), but that is not enough for complex anisotropic elastoplastic deformations.
3. Recently, the fixed-stress split has become a popular method to integrate a solid mechanics finite element solver into existing reservoir simulators that use TPFA for numerical discretization (Almani et al., 2016; Castelletto et al., 2015; Jha and Juanes, 2014; Kim et al., 2011). This is a big improvement regarding the previous point. However, while this approach is well-suited for isotropic elasticity, it may lead to excessive nested Newton iterations when applied to complex anisotropic elastoplastic models.

The second limitation on the numerical discretization approach follows directly from the third point in the previous list. We all know the standard mixed finite element method (FEM) is a perfect candidate to handle hydromechanical coupling, anisotropy, and elastoplasticity, and it usually adopts the fully implicit coupling strategy, which is unconditionally stable and highly accurate (Almani et al., 2016; Tang et al., 2024; Wang and Song, 2020; Wu et al., 2025). However, most hydromechanical FEM simulator only considers an incompressible or a slightly compressible fluid, the incorporation of a compressible gas as well as the associated numerical ramifications are rarely mentioned. Although FEM-based commercial software COMSOL can choose the fluid type, few studies have provided a complete finite element framework for fractured shale gas reservoirs, nor demonstrated how to couple gas flow in fractures with the flow in the rock matrix. The only known attempt thus far to propose a finite element framework is by Lu et al. (2021), although they ultimately relied on COMSOL rather than an in-house simulator. Moreover, their final outcome neither presents the true weak form format nor provides the finite element residual vector and Jacobian matrix. Their discretization of hydraulic fractures is not comprehensive, and the treatment of mass transfer between the fracture and the matrix is not discussed. Therefore, incorporation of a compressible gas into the current hydromechanical FEM formulation is indeed necessary and will be beneficial for future modelers.

Another "obvious" yet rarely mentioned paradox in literature concerns the "geomechanics effect on gas production". Although the term "geomechanics" is not precise since fluid flow through porous media also belongs to the subject of geomechanics, nowadays, this term is used to describe that permeability changes with stress or deformation. Yu and Sepehrnoori (2014) is one of the earliest works in the literature that discussed the geomechanics effect. In that work, the geomechanics implies the effect of closure stress on fracture conductivity, and it has a negative effect on the cumulative gas production. Later, other studies incorporated the solid deformation module and got similar conclusions (Jiang and Yang, 2018; Micheal et al., 2022; Yan et al., 2018a; Zhao et al., 2020). However, there is an obvious contradiction: incorporating a solid deformation module implies an additional compression of the pore space, compared to the reference case of rigid porous skeleton and constant porosity. This is particularly true in scenarios involving elastoplastic deformation or a significant reduction in the Young's modulus of the porous matrix. Such conditions should enhance the ultimate gas production amount, given the same initial and final gas pressures. In other words, even though geomechanics undoubtedly reduces permeability, in the generated figure of cumulative gas production vs. time, we would still expect the scenario that incorporates geomechanics to eventually approach and surpass the scenario that does not. Unfortunately, none of the existing studies have shown this trend. Consequently, we are curious whether, in numerical simulations, extending the total simulation time, even if it goes far beyond the usual 30 year horizon found in the literature, would reveal this behavior. Addressing this research gap is of paramount importance to ensure our basic intuition of poromechanics still works.

To accomplish the aforementioned goals, this paper is organized to offer a thorough analysis. We start in Section 2 with the mathematical formulation, establishing the foundation with finite element equations for both the porous matrix and the fracture flow model. Section 3 verifies our approach through a regular fracture network and a strip load problem. We then move to Section 4, where we have a very comprehensive and thorough discussion on various factors that would affect the gas production, and in particular, the bedding plane orientation and the geomechanics effect. Section 5 wraps up our findings, summarizing the contributions to the field of unconventional energy geomechanics and their practical implications.

2. Mathematical formulation

2.1. Finite element equations for the porous matrix

In porous matrix, the coupled gas flow and solid deformation equations fit nicely into the standard finite element framework, with solid displacement vector $\boldsymbol{u}$ and gas pressure p as primary unknowns. In this section, we briefly review the $\boldsymbol{u}/p$ formulation of the boundary-value problem following the lines of Zhao and Borja (2020), and we only highlight the salient features of the formulation with an emphasis on how to handle a highly compressible fluid. The relevant field equations ignoring gravity and adsorbed gas mass on domain Ω include the balance of linear momentum,

$$\nabla \cdot \left(\boldsymbol{\sigma}' - p\,\boldsymbol{b}\right) = \boldsymbol{0}\,, \tag{1}$$

and the balance of mass for gas flow,

$$\frac{1}{m}\frac{\mathrm{d}p}{\mathrm{d}t} + \boldsymbol{b} : \frac{\mathrm{d}\boldsymbol{\epsilon}}{\mathrm{d}t} + \frac{1}{K_g}\boldsymbol{q}\cdot\nabla p + \nabla\cdot\boldsymbol{q} = 0\,, \tag{2}$$

where in the first Eq. (1), $\boldsymbol{\sigma}' = \mathbb{C}^e : \boldsymbol{\epsilon}^e = \mathbb{C}^e : (\boldsymbol{\epsilon} - \boldsymbol{\epsilon}^p)$ is the effective stress tensor, $\mathbb{C}^e$ is the fourth-order elastic tensor, $\boldsymbol{\epsilon}^e$ is the elastic strain tensor, $\boldsymbol{\epsilon}^p$ is the plastic strain tensor, p is the gas pressure, and $\boldsymbol{b}$ is the Biot tensor. According to the theory of elasticity, for anisotropic material, we have the following expression for $\mathbb{C}^e$ (Zhang and Borja, 2021)

$$\begin{aligned}\mathbb{C}^e =& \lambda^e \mathbf{1}\otimes\mathbf{1} + 2\mu_T\mathbb{I} + a^e\left(\mathbf{1}\otimes\boldsymbol{M} + \boldsymbol{M}\otimes\mathbf{1}\right)\\ &+ b^e\boldsymbol{M}\otimes\boldsymbol{M} + 2\left(\mu_L - \mu_T\right)\left(\mathbf{1}\odot\boldsymbol{M} + \boldsymbol{M}\odot\mathbf{1}\right),\end{aligned} \tag{3}$$

where $\mathbf{1}$ is the second-order identity tensor, $\mathbb{I}$ is the symmetric fourth-order identity tensor, the tensorial operator $\odot$ is defined such that $(\boldsymbol{A}\odot\boldsymbol{B})_{ijkl} = (A_{ik}B_{jl} + A_{il}B_{jk})/2$, $\boldsymbol{M}$ is known as the micro-structure tensor that has the definition $\boldsymbol{M} = \boldsymbol{n}\otimes\boldsymbol{n}$, and $\boldsymbol{n}$ is the unit normal vector of the bedding plane. Instead of two Lamé constants in isotropic elasticity, Eq. (3) contains five material parameters λ^e, μ_T, a^e, b^e, and μ_L that characterize different mechanical behaviors between the bed-normal

(BN) and bed-parallel (BP) directions. By using the poromechanics theory, the Biot tensor $\boldsymbol{b}$ is constructed as (Cheng, 2016)

$$\boldsymbol{b} = \mathbf{1} - \frac{\mathbb{C}^e : \mathbf{1}}{3K_s}, \tag{4}$$

where K_s is the bulk modulus of the solid grain. If we let K_s approach infinity, we recover the familiar isotropic Biot coefficient that is equal to 1.

As for notations in Eq. (2), $\boldsymbol{\epsilon} = \mathrm{sym}(\nabla \boldsymbol{u})$ is the total compatible strain tensor, m is the generalized Biot modulus that has the following expression

$$\frac{1}{m} = \frac{\phi}{K_g} + \frac{1}{K_s}\left(1 - \phi - \frac{\mathbf{1} : \mathbb{C}^e : \mathbf{1}}{9K_s}\right), \tag{5}$$

in which ϕ is the porosity, K_g is the gas bulk modulus, $\boldsymbol{q}$ is the gas flow velocity defined by $-\boldsymbol{\kappa} \cdot \nabla p$, and $\boldsymbol{\kappa}$ is the apparent mobility tensor, which is the apparent permeability divided by gas viscosity μ_g. Different from the slightly compressible fluid such as water, for gas, the aforementioned parameters m, K_g, and $\boldsymbol{\kappa}$ are all pressure sensitive, *i.e.*, they are functions of gas pressure p (Yan et al., 2018a). For ideal gas, K_g is equal to the gas pressure p. These characteristics would bring more non-linearities to the expressions of the tangent operator, as will be shown in the following paragraphs. It is worth mentioning that these conservation laws must be combined with relevant proper initial and boundary conditions to solve for the unknown fields (Zhao and Borja, 2020).

To derive the corresponding variational equations (weak form), two sets of weighting functions $\boldsymbol{\eta}$ and φ are defined such that the following equations are satisfied:

$$\int_\Omega \nabla^s \boldsymbol{\eta} : \left(\boldsymbol{\sigma}' - p\,\boldsymbol{b}\right) \mathrm{d}V = \int_{\partial\Omega_t} \boldsymbol{\eta} \cdot \hat{\boldsymbol{t}}\, \mathrm{d}A, \tag{6}$$

$$\int_\Omega \varphi \left(\frac{1}{m}\frac{\mathrm{d}p}{\mathrm{d}t} + \boldsymbol{b} : \frac{\mathrm{d}\boldsymbol{\epsilon}}{\mathrm{d}t} + \frac{1}{K_g}\boldsymbol{q} \cdot \nabla p\right) \mathrm{d}V - \int_\Omega \nabla\varphi \cdot \boldsymbol{q}\, \mathrm{d}V = \int_{\partial\Omega_q} \varphi\, \hat{q}\, \mathrm{d}A, \tag{7}$$

where ∇^s denotes the symmetric gradient operator for a vector field, $\hat{\boldsymbol{t}}$ is the prescribed traction vector on boundary $\partial\Omega_t$, and $\hat{q}$ is the prescribed inward fluid flux (scalar) on boundary $\partial\Omega_q$. Next, we substitute the trial and weighting functions with their finite-dimensional Galerkin approximations $\boldsymbol{u}^h$, p^h, $\boldsymbol{\eta}^h$, and φ^h that are constructed using standard shape functions, and for the rate dependence embedded in Eq. (7), the first-order accurate, unconditionally stable backward difference scheme is employed to integrate Eq. (7) from the previous time step t_n to the current time step t_{n+1} (Zhang et al., 2019a). The resulting time-integrated flow equation takes the form:

$$\begin{aligned}&\int_\Omega \varphi^h \left[\frac{p^h - p_n^h}{m^h} + \boldsymbol{b} : \left(\boldsymbol{\epsilon}^h - \boldsymbol{\epsilon}_n^h\right) + \frac{\Delta t}{K_g^h}\boldsymbol{q}^h \cdot \nabla p^h\right] \mathrm{d}V \\ &- \Delta t \int_\Omega \nabla\varphi^h \cdot \boldsymbol{q}^h\, \mathrm{d}V = \Delta t \int_{\partial\Omega_q} \varphi^h\, \hat{q}^h\, \mathrm{d}A,\end{aligned} \tag{8}$$

where p_n^h and $\boldsymbol{\epsilon}_n^h$ are the gas pressure and strain fields evaluated at time t_n, respectively, $\Delta t = t_{n+1} - t_n$ is the time increment, and all the other quantities without any subscript are evaluated at time step t_{n+1}. Note that different from the flow equation for slightly compressible fluid, in Eq. (8), m and K_g are also evaluated implicitly by using p^h. For Eq. (6), all the Galerkin approximations are simply evaluated at t_{n+1} without any time integration. Finally, the Galerkin form is converted to the residual matrix form as follows (we omit the superscript h for brevity)

$$\mathcal{R}_u = \int_\Omega \mathbb{B}^\top \left\{\boldsymbol{\sigma}' - p\,\boldsymbol{b}\right\} \mathrm{d}V - \int_{\partial\Omega_t} \mathbb{N}_u^\top \hat{\boldsymbol{t}}\, \mathrm{d}A, \tag{9}$$

$$\begin{aligned}\mathcal{R}_p = &\int_\Omega \mathbb{N}_p^\top \left[\frac{p - p_n}{m} + \boldsymbol{b} : \left(\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_n\right) + \frac{\Delta t}{K_g}\boldsymbol{q} \cdot \nabla p\right] \mathrm{d}V \\ &- \Delta t \int_\Omega \mathbb{E}^\top \boldsymbol{q}\, \mathrm{d}V - \Delta t \int_{\partial\Omega_q} \mathbb{N}_p^\top\, \hat{q}\, \mathrm{d}A,\end{aligned} \tag{10}$$

where $\mathbb{N}$ denotes the shape function matrix for either the displacement field (with subscript u) or the pressure field (with subscript p), $\mathbb{B} = \nabla^s \mathbb{N}_u$ is the strain–displacement transformation matrix, $\mathbb{E} = \nabla \mathbb{N}_p$ is the gradient-pressure transformation matrix. Here, we remark that $\{\boldsymbol{\sigma}' - p\,\boldsymbol{b}\}$ is the total stress tensor written in Voigt form. Due to the gas compressibility and possible non-Darcy flow (Zhang et al., 2024), the residual vectors defined above are nonlinear with respect to the nodal unknown vectors $\boldsymbol{d}$ and $\boldsymbol{p}$, and therefore, they are solved by the Newton–Raphson iteration scheme. The iteration requires the evaluation of consistent tangent operator, which takes the form (Zhao and Borja, 2020):

$$\begin{bmatrix} \mathcal{K}_{11} & \mathcal{K}_{12} \\ -\mathcal{K}_{12}^\top & \mathcal{K}_{22} \end{bmatrix} \begin{pmatrix} \delta\boldsymbol{d} \\ \delta\boldsymbol{p} \end{pmatrix} = -\begin{pmatrix} \mathcal{R}_u \\ \mathcal{R}_p \end{pmatrix}. \tag{11}$$

The individual matrices comprising the tangent operator are given as follows:

$$\mathcal{K}_{11} = \int_\Omega \mathbb{B}^\top \left\{\mathbb{C}^{ep}\right\} \mathbb{B}\, \mathrm{d}V, \tag{12}$$

$$\mathcal{K}_{12} = -\int_\Omega \mathbb{B}^\top \left\{\boldsymbol{b}\right\} \mathbb{N}_p\, \mathrm{d}V, \tag{13}$$

$$\begin{aligned}\mathcal{K}_{22} = &\Delta t \int_\Omega \left[\mathbb{E}^\top - \mathbb{N}_p^\top \frac{(\nabla p)^\top}{K_g}\right] \left[\left(\frac{\partial \boldsymbol{\kappa}}{\partial p} \cdot \nabla p\right) \mathbb{N}_p + \boldsymbol{\kappa}\, \mathbb{E}\right] \mathrm{d}V \\ &+ \int_\Omega \mathbb{N}_p^\top \frac{\partial}{\partial p}\left(\frac{p - p_n}{m}\right) \mathbb{N}_p\, \mathrm{d}V + \Delta t \int_\Omega \mathbb{N}_p^\top \boldsymbol{q}^\top \left[\frac{\mathbb{E}}{K_g} - \left(\frac{1}{K_g^2}\frac{\partial K_g}{\partial p}\nabla p\right) \mathbb{N}_p\right] \mathrm{d}V,\end{aligned} \tag{14}$$

where $\mathbb{C}^{ep}$ is the fourth-order algorithmic elastoplastic tensor (Borja, 2013), ∇p implies a column vector of the space dimension, and $\{\circ\}$ represents the Voigt notation. Note that the calculation of $\partial\boldsymbol{\kappa}/\partial p$ and $\partial m/\partial p$ could be evaluated numerically for complicated apparent permeability models (Civan, 2010; Darabi et al., 2012; Freeman et al., 2011), gas viscosity models, and gas pressure–volume–temperature (PVT) relationships (Zuo et al., 2018), and we ignore the derivative of $\boldsymbol{\kappa}$ with respect to $\boldsymbol{\epsilon}$, otherwise Eq. (14) cannot be derived in such an analytical form.

Remark 2.1. It is worthwhile to discuss the block tangent operator of Eq. (11) in the undrained limit ($\Delta t \to 0$), in comparison to that in White and Borja (2008). In the undrained limit, $\mathcal{K}_{22}$ is simplified from Eq. (14) as

$$\mathcal{K}_{22} = \int_\Omega \mathbb{N}_p^\top \frac{\partial}{\partial p}\left(\frac{p - p_0}{m}\right)\Big|_{p=p_0} \mathbb{N}_p\, \mathrm{d}V = \int_\Omega \frac{\mathbb{N}_p^\top \mathbb{N}_p}{m}\, \mathrm{d}V, \tag{15}$$

where p_0 is the initial gas pressure as $\Delta t \to 0$. For ideal gas, we know $1/m \geq \phi/K_g = \phi/p > 0$, so $\mathcal{K}_{22} \succ 0$ is a symmetric positive definite matrix. Therefore, the equal-order linear interpolation for displacement and gas pressure fields would not cause spurious pressure oscillation in the undrained limit, which is different from the case of incompressible fluid in White and Borja (2008). As a result, the Polynomial-Pressure-Projection (PPP) correction term is not strictly necessary in the coupled gas flow and solid deformation analysis; though including it only yields a negligible difference in the results. This is because the solid shear modulus $\bar{G}$ in the PPP term is usually much larger than the gas pressure p, so $\tau/(2\bar{G})$ is much less than $1/m$, in which τ is a stabilization parameter and it is usually on the order of one (White and Borja, 2008).

Remark 2.2. For compressible fluid including gas, a non-zero $\mathcal{K}_{22}$ would lead to an overshoot in pressure near the boundary (higher than the initial pressure) as $\Delta t \to 0$, which is briefly explained here as follows using the 1D analogy.

Suppose we have five uniform one-dimensional (1D) linear elements, each of length $l^e = 1$ (for a total of six nodes). We can compute the element stiffness matrix k_{22}^e in the undrained limit and the element

stabilization matrix k^e_{22s} in the PPP method following White and Borja (2008):

$$k^e_{22} = \frac{1}{m}\begin{bmatrix}1/3 & 1/6\\ 1/6 & 1/3\end{bmatrix}, \qquad k^e_{22s} = \frac{\tau}{2\bar{G}}\begin{bmatrix}1/12 & -1/12\\ -1/12 & 1/12\end{bmatrix}. \tag{16}$$

Here, we assume constant m and $\tau/(2\bar{G})$ across all elements. Without loss of generality, let $1/m = \tau/(2\bar{G}) = 1$. We then assemble the element matrix contributions into the corresponding 6×6 global matrices, denoted by $\mathcal{K}^{\mathrm{u}}_{22}$ and $\mathcal{K}_{22s}$, respectively (where the superscript "u" indicates the undrained limit). Initially, all six nodes have a pressure of 1. At the very beginning ($t = 0^+$), the pressure at the first node instantaneously drops to 0.1. To determine the pressure distribution in the undrained limit for a conduction problem (ignoring $\mathcal{K}_{12}$), we solve

$$\mathcal{K}^{\mathrm{u}}_{22}\,\Delta p = 0, \tag{17}$$

subject to the constraint at the first node

$$\Delta p(1) = 0.1 - 1 = -0.9\,. \tag{18}$$

Here, Δp represents the change in pressure from the initial state. Eq. (18) implies that the first row of Eq. (17) does not necessarily hold, otherwise there is only one trivial solution of $\Delta p = 0$ since K^{u}_{22} is invertible. The solution is

$$p^{\mathrm{u}} = \left[0.1,\quad 1.2415,\quad 0.9354,\quad 1.0174,\quad 0.995,\quad 1.0025\right]^{\top}, \tag{19}$$

from which we observe that p^{u} can exceed 1 at certain nodes. Now if we replace $\mathcal{K}^{\mathrm{u}}_{22}$ with $\mathcal{K}_{22s}$, we find all the nodes would have the same value as the node 1, *i.e.*, $p^{\mathrm{u}} = 0.1$, which is not physically correct because the steady-state cannot be reached immediately for such a system. In other words, for porous media saturated with highly compressible fluid, an improper stabilization parameter can severely compromise solution accuracy.

Remark 2.3. When we have discrete fractures embedded in the matrix domain, the last term in the expression of $\mathcal{R}_p$ in Eq. (10) should be modified to consider the mass transfer between the fracture and the matrix $\hat{q}_{mf}$ as

$$\mathcal{R}^{\mathrm{new}}_p = \mathcal{R}^{\mathrm{Eq.(10)}}_p + \Delta t \int_{\bar{\Omega}_F} \mathbb{N}^{\top}_p\, \hat{q}_{mf}\, \mathrm{d}A\,, \tag{20}$$

where $\hat{q}_{mf} > 0$ implies the flow from the porous matrix to the fracture, so that is why we have the plus sign in the above equation.

2.2. Finite element equations of the fracture flow model

We consider the 1D fracture domain $\bar{\Omega}_F$, as shown in Fig. 1, which has one starting point and one ending point, and they are regarded as the boundary of $\bar{\Omega}_F$, denoted as $\partial\bar{\Omega}_F = \{\mathrm{Point}_{start},\ \mathrm{Point}_{end}\}$. In the fracture flow model, we assume the pressure continuity at the fracture-matrix interface, and the discrete fracture elements must be located on the edges of porous matrix elements, sharing the same nodes (Chen et al., 2020; Watanabe et al., 2012). For $\partial\bar{\Omega}_F$, we assume either a zero-flux boundary condition or a Dirichlet boundary condition is prescribed (Jin and Arson, 2020; Shi et al., 2022).

On $\bar{\Omega}_F$, a local coordinate system denoted by ζ axis is built, and the strong form of the fracture flow equation is given as (Chen et al., 2020)

$$\frac{\partial\left(w\phi_F\rho_g\right)}{\partial t} + \frac{\partial\left(\rho_g w q_F\right)}{\partial\zeta} = \rho_g\,\hat{q}_{mf}\,, \tag{21}$$

$$q_F = -\frac{k_F}{\mu_g}\frac{\partial p_F}{\partial\zeta}\,, \tag{22}$$

where w is the fracture aperture, ϕ_F is the fracture porosity, ρ_g is the gas density, μ_g is the gas viscosity, and k_F is the fracture permeability. Note that we use p_F to emphasize the fracture flow, but it is just the matrix gas pressure p at the fracture-matrix interface including the shared nodes.

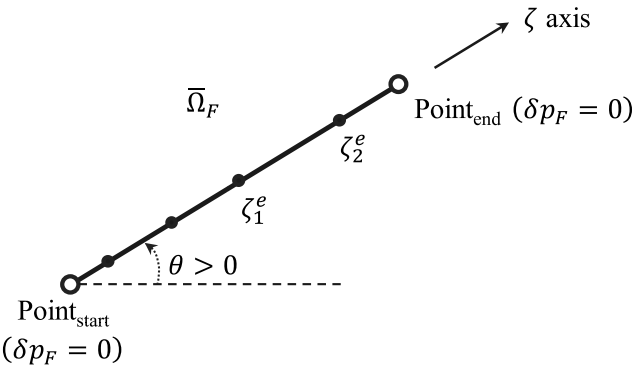


Fig. 1. A schematic demonstration of the 1D fracture domain $\bar{\Omega}_F$. The domain has one starting point and one ending point, and they are regarded as the boundary of $\bar{\Omega}_F$. Please refer to Appendix B of Lu et al. (2021) for a similar demonstration. Please note that $\delta p_F = 0$ only applies to the point if its pressure is known. For zero-flux condition ($q_F = 0$) or if this point locates inside the whole fracture, $\delta p_F = 0$ does not apply anymore.

Further simplifications of Eq. (21) are needed before the derivation of the weak form. In this work, ϕ_F and w are assumed to be either constant or they are updated explicitly after the convergence of each time step. As a result, Eq. (21) is rewritten as

$$w\phi_F\frac{\partial\rho_g}{\partial t} + w\frac{\partial\rho_g}{\partial\zeta}q_F + w\rho_g\frac{\partial q_F}{\partial\zeta} = \rho_g\,\hat{q}_{mf}\,. \tag{23}$$

Then we divide ρ_g on both sides and by using the definition of K_g, we obtain another equivalent form as

$$\frac{w\phi_F}{K_g}\frac{\partial p_F}{\partial t} + \frac{w}{K_g}\frac{\partial p_F}{\partial\zeta}q_F + w\frac{\partial q_F}{\partial\zeta} = \hat{q}_{mf}\,. \tag{24}$$

To obtain the weak form corresponding to Eq. (24), we multiply Eq. (24) by the weighting function δp_F and through integration by part, we obtain

$$\int_{\bar{\Omega}_F}(\delta p_F)\frac{w\phi_F}{K_g}\frac{\partial p_F}{\partial t}\,\mathrm{d}\zeta + \int_{\bar{\Omega}_F} w q_F\left[\frac{\delta p_F}{K_g}\frac{\partial p_F}{\partial\zeta} - \frac{\partial(\delta p_F)}{\partial\zeta}\right]\mathrm{d}\zeta - \int_{\bar{\Omega}_F}(\delta p_F)\,\hat{q}_{mf}\,\mathrm{d}\zeta = 0\,. \tag{25}$$

In the above simplification to Eq. (25), we have used the fact that either $q_F = 0$ or $\delta p_F = 0$ holds on $\partial\bar{\Omega}_F$. Next, by adopting the backward Euler time integration scheme, $\partial p_F/\partial t$ is approximated as $\left(p_F - p_F^{t_n}\right)/\Delta t$, where the superscript t_n implies the quantity at the previous time step, other quantities are by default evaluated implicitly. Note that the fracture aperture w is introduced to ensure unit consistency with the finite element equation of the porous matrix.

In this work, we assume a 1D linear element which means that for one element with local coordinates ζ^e_1 and ζ^e_2, the shape function $[N^e_F]$ and its derivative in the local coordinate system $[E^e_F]$ are

$$[N^e_F] = \left[\frac{\zeta - \zeta^e_2}{\zeta^e_1 - \zeta^e_2},\quad \frac{\zeta^e_1 - \zeta}{\zeta^e_1 - \zeta^e_2}\right], \quad [E^e_F] = \left[\frac{1}{\zeta^e_1 - \zeta^e_2},\quad \frac{-1}{\zeta^e_1 - \zeta^e_2}\right]. \tag{26}$$

Through the assembly operation of the element shape functions and derivatives, the global matrix equation in the residual form is given as

$$\begin{aligned}\mathcal{R}_F = &\int_{\bar{\Omega}_F}[N_F]^{\top}\frac{w\phi_F}{K_g}\left(p_F - p_F^{t_n}\right)\mathrm{d}\zeta + \Delta t\int_{\bar{\Omega}_F}[N_F]^{\top}\frac{w}{K_g}\frac{\partial p_F}{\partial\zeta}q_F\,\mathrm{d}\zeta\\ &- \Delta t\int_{\bar{\Omega}_F}[E_F]^{\top}w\,q_F\,\mathrm{d}\zeta - \Delta t\int_{\bar{\Omega}_F}[N_F]^{\top}\hat{q}_{mf}\,\mathrm{d}\zeta\,.\end{aligned} \tag{27}$$

By using the global $[N_F]$ and $[E_F]$ matrices, we have $p_F = [N_F][p_F]$ and $\partial p_F/\partial\zeta = [E_F][p_F]$. The size of $[N_F]$ (row vector), $[E_F]$ (row vector), and $[p_F]$ (column vector) is equal to the number of nodes in $\bar{\Omega}_F$. The residual $\mathcal{R}_F$ is generally nonlinear with respect to the unknown nodal

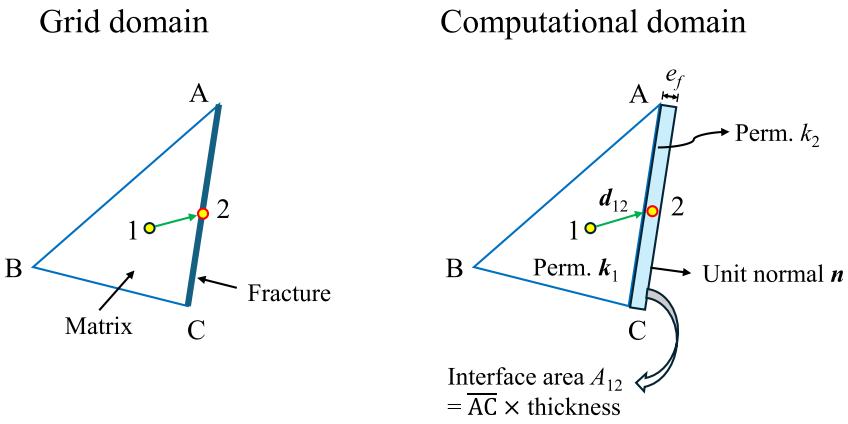


Fig. 2. Schematic diagram of the transmissibility evaluation for connection between matrix and fracture control volumes.

pressure vector $[p_F]$ and is best solved using Newton's method. As a result, the algorithmic tangent operator $\mathcal{K}_F$ is given as

$$\begin{aligned}\mathcal{K}_F = &\int_{\bar{\Omega}_F} [N_F]^\top w \phi_F \left[\frac{\partial c_g}{\partial p}\left(p_F - p_F^{t_n}\right) + c_g\right][N_F]\,\mathrm{d}\zeta \\ &+ \Delta t \int_{\bar{\Omega}_F} [N_F]^\top w \frac{\partial c_g}{\partial p}\frac{\partial p_F}{\partial \zeta} q_F [N_F]\,\mathrm{d}\zeta - \Delta t \int_{\bar{\Omega}_F} [N_F]^\top w\, c_g\, \chi_F [E_F]\,\mathrm{d}\zeta \\ &+ \Delta t \int_{\bar{\Omega}_F} [E_F]^\top w \frac{k_F}{\mu_g}[E_F]\,\mathrm{d}\zeta\,,\end{aligned} \tag{28}$$

where

$$\chi_F = \frac{\partial p_F}{\partial \zeta}\frac{k_F}{\mu_g} - q_F \tag{29}$$

and we have introduced the notation of gas compressibility $c_g = 1/K_g$. Note that the $\mathcal{R}_F$ and $\mathcal{K}_F$ should be superimposed to the finite element equations of the porous matrix Eq. (11), in order to satisfy the solvability requirements (Chen et al., 2020), and an illustrative sketch is given in `Figure 4` of Segura and Carol (2004) and `Figure 7` of Chen et al. (2020).

Remark 2.4. It is important to discuss the mass transfer/exchange term $\hat{q}_{mf}$ between the fracture and the matrix. Through superimposition, the terms for flux exchange between the two systems balance off and cancel internally (note that $\mathbb{N}_p = [N_F]$ on $\bar{\Omega}_F$), so its explicit calculation is not required. This point has been precisely mentioned by Woodbury and Zhang (2001), Segura and Carol (2004), and Watanabe et al. (2012). Note that in some literature, authors have simply omitted $\hat{q}_{mf}$ in Eq. (21), yet arrived at the same formulation coincidentally (Fan et al., 2015; Li et al., 2020; Liu et al., 2019; Zhao et al., 2020). However, it is crucial to understand the underlying reasoning in this scenario.

Nevertheless, this does not imply that we cannot evaluate the mass transfer after obtaining the solution at all nodes. The evaluation adopts the methodology of Two-Point Flux Approximation (TPFA), as shown in the following Fig. 2. We calculate the mass transfer Q_{12}, which is actually the volumetric flow rate under standard conditions, using the following equation:

$$Q_{12} = T_{12}\,\lambda\,(P_1 - P_2), \qquad P_1 = \frac{p_A + p_B + p_C}{3}, \qquad P_2 = \frac{p_A + p_C}{2}, \tag{30}$$

where T_{12} represents the geometric part of the transmissibility and $\lambda = 1/(\mu_g B_g)$ is the fluid part of the transmissibility (Karimi-Fard and Firoozabadi, 2003). In the definition of λ, B_g is known as the gas formation volume factor. As shown by the second and third subequations of Eq. (30), P_1 could be obtained from the average of nodal pressures of one triangle element, and P_2 could be obtained from the average of nodal pressures of one fracture element. In TPFA, the evaluation of λ adopts the upwinding strategy, while the evaluation of T_{12} requires the harmonic average and the geometric information of the fracture as

$$T_{12} = \frac{\alpha_1\alpha_2}{\alpha_1 + \alpha_2}, \qquad \alpha_1 = \frac{\left|\boldsymbol{d}_{12}\cdot(\boldsymbol{k}_1\cdot\boldsymbol{n})\right| A_{12}}{\boldsymbol{d}_{12}\cdot\boldsymbol{d}_{12}}, \qquad \alpha_2 = A_{12}\frac{2k_2}{e_f}, \tag{31}$$

where A_{12} is the interface area, $\boldsymbol{d}_{12}$ is a vector that connects the element centroid and the fracture middle point, $\boldsymbol{n}$ is the unit normal vector of the interface, e_f is the fracture aperture, $\boldsymbol{k}_1$ represents the matrix permeability tensor, and k_2 represents the fracture permeability.

Remark 2.5. As our last remark in this section, we briefly discuss the displacement field across the fracture. In the current formulation, we do not consider the displacement discontinuity across the fracture, but in the benchmark test of the published open source code, this is considered through the zero-thickness interface element (Jha and Juanes, 2014; Jiang and Yang, 2018; Segura and Carol, 2004). In literature, this displacement discontinuity could be handled by many other methods such as the node to surface or surface to surface contact (Peng et al., 2025; Wang et al., 2022b), extended finite element method (XFEM) (Yan et al., 2018a), assumed enhanced strain finite element method (AES-FEM) (Zhao et al., 2025), and the phase field method (Fei et al., 2022; Xu et al., 2024; Yuan et al., 2025).

Nevertheless, we did not consider this issue in the subsequent model application for two reasons. First, our model application already includes an elastoplastic constitutive law. If we further account for the displacement discontinuity across the fracture, the stress singularities near the fracture tips could easily cause the algorithm to fail to converge. Second, even without applying the zero-thickness interface element, we can see from Fig. 15 and Fig. 23 that there is a sharp horizontal displacement transition across the fracture, which achieves the intended effect.

3. Model verification

In this section, we verify the fracture flow model and the hydromechanical coupling of the porous matrix using a regular fracture network and a strip load problem, respectively. All simulations are conducted under plane strain conditions, and gravity is ignored.

Although the mathematical framework developed in Section 2 was formulated for ideal gas, it is actually highly general. Substituting a slightly compressible fluid or incompressible fluid in place of the ideal gas requires virtually no modifications. In fact, one only need to change the expressions for K_g and $1/m$ in the simulator. Therefore, in the following model verification examples, the fluid could be either water or ideal gas.

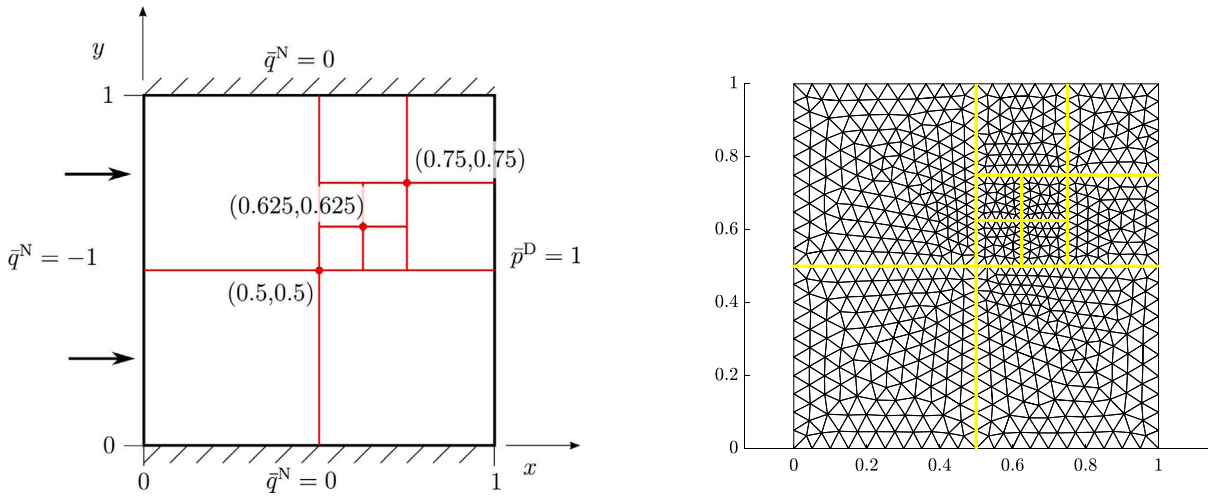


Fig. 3. Schematic diagram of the regular fracture network: geometry, boundary conditions, and the finite element mesh. Note the fractures are discretized using one-dimensional elements, as already indicated by Fig. 1.

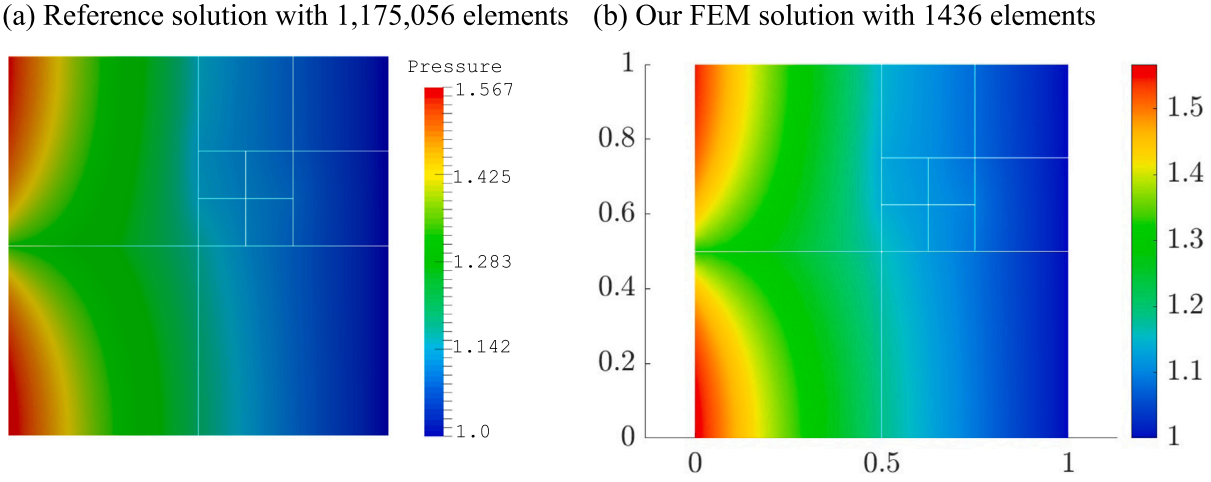


Fig. 4. Comparison of our FEM solution with the reference solution obtained from an extremely fine mesh. The two pressure distributions throughout the domain are visually indistinguishable.

3.1. Regular conductive fracture network

Fig. 3 presents the computational domain, including the fracture network, finite element mesh, and boundary conditions. The fluid is assumed to be incompressible. The matrix permeability is isotropic ($k_m = 1$), while all fractures share a uniform aperture of 10^{-4} and a permeability of $k_F = 10^4$. Because this setup is purely hypothetical for model verification, we do not specify the length unit. We set the Biot tensor $\boldsymbol{b}$ to $\mathbf{0}$, which ensures $\mathcal{K}_{12} = 0$ from Eq. (13), thereby decoupling fluid flow from solid deformation. For the solid mechanics aspect, two corner nodes on the bottom edge are constrained, like a simply supported beam, to eliminate rigid body motion. As a result, displacement remains at $\mathbf{0}$, consistent with our goal of focusing solely on the fluid pressure distribution. The time step setting is trivial because we are solving a steady-state problem. Additionally, the value of ϕ_F is irrelevant and can take any numbers.

Figs. 4 and 5 illustrate the comparison results between our FEM solution and the reference solution across the entire domain and along two specific lines, respectively. The reference solution is obtained from an extremely fine mesh using MFD, which resolves every fracture with 10 elements in its normal direction and becomes coarser away from the fractures (Flemisch et al., 2018). In either figure, the match is excellent and visually indistinguishable. Therefore, following a similar route in Flemisch et al. (2018), we choose to calculate the solution error, which provides a quantitative measure of the goodness of our FEM solution. The method for such error calculation is documented comprehensively in this Git repository of Flemisch et al. (2018). The results are summarized in Table 1. By comparing with the existing results obtained using other advanced finite volume or XFEM-based methods in `Table 3` and `Table 4` of Flemisch et al. (2018), our method is not only comparable to the best existing numerical methods but also outperforms them in certain aspects. In other words, although FEM is frequently criticized for the issue of local mass conservation, its performance for single-phase flow remains reliable and can lead to satisfactory results.

Next, we alter the fluid type to ideal gas for an additional verification. While no reference solution exists in the literature for ideal gas with the same discrete fracture network, we can simplify the problem by eliminating fracture contributions. This leaves only gas flow through the porous matrix, allowing us to derive an analytical solution, as shown below:

$$p_g^{\text{analytical}} = \sqrt{3 + 2\sqrt{2} - (2 + 2\sqrt{2})x}\,. \tag{32}$$

This solution differs significantly from the analytical solution for incompressible fluid, which is $p_w^{\text{analytical}} = 2 - x$. The verification results

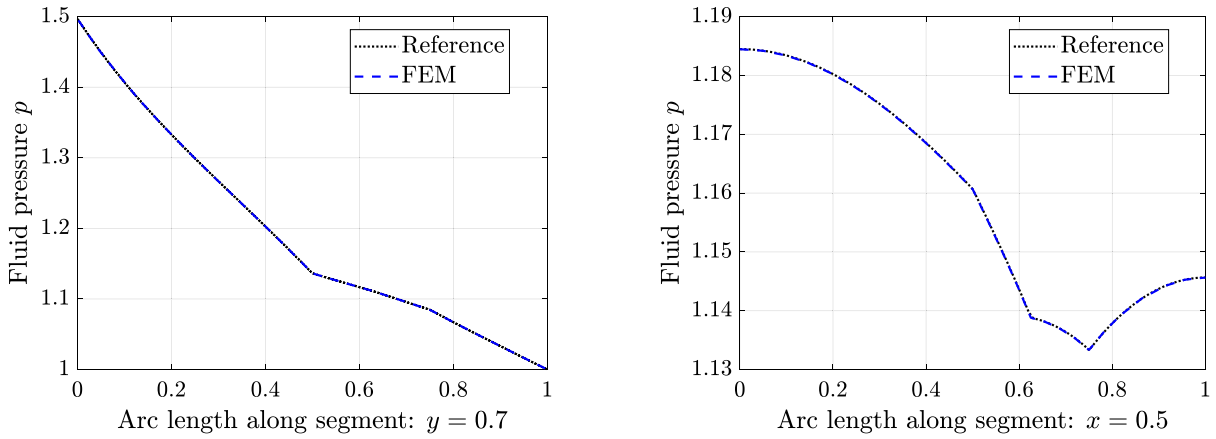


Fig. 5. Comparison of two solutions by plotting along two lines. Again, the results appear nearly identical. Therefore, we choose to calculate the solution error and $\mathcal{K}_{22}$ matrix characteristics quantitatively, as shown in Table 1.

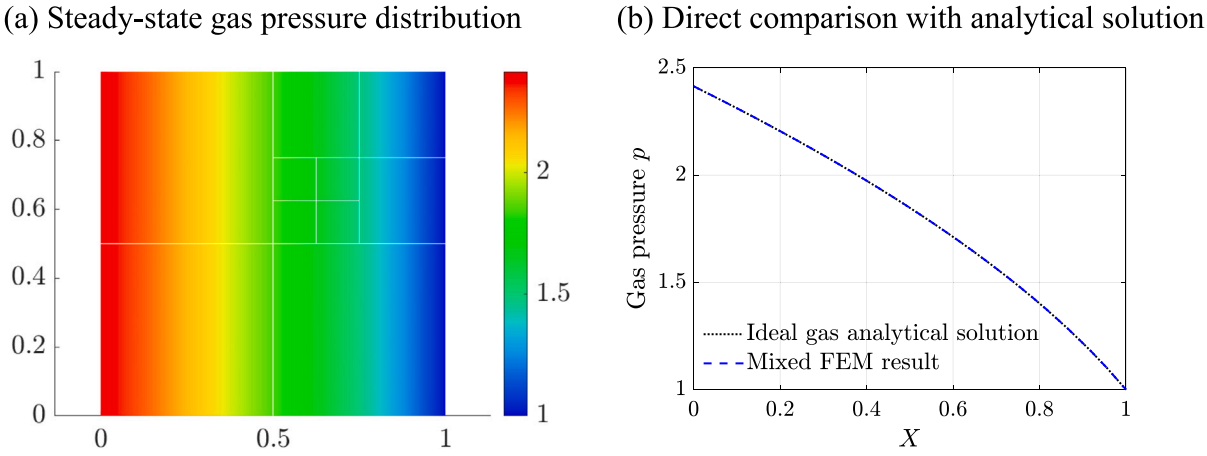


Fig. 6. By replacing the previously considered incompressible fluid with a compressible ideal gas and neglecting any fracture contributions, we derive an analytical expression for the gas pressure, enabling an additional comparison with our numerical results. The agreement between the analytical and numerical solutions remains excellent.

Table 1
Quantitative solution errors and $\mathcal{K}_{22}$ matrix characteristics for the fracture network problem. In the table header, E_{ma} represents the error computed from the centroid solution of triangle elements, and E_{fr} represents the error computed from the ending points solution of fracture segments. Matrix sparsity is calculated as the portion of non-zero numbers in $\mathcal{K}_{22}$. The values reported in this table should be compared with those values in `Table 3` and `Table 4` of Flemisch et al. (2018).

Case	E_{ma}	E_{fr}	Matrix sparsity	Condition number
Conductive fracture	0.0098	0.0085	0.0089	2.1951×10^4
Blocking fracture	0.0037	0.0057	0.0079	5.0443×10^3

are illustrated in Fig. 6, and once again, the agreement between the analytical and numerical solutions is excellent.

As a side note, Appendix A presents the comparison results for the blocking fracture network, *i.e.*, the fractures are assigned a permeability much lower than that of the porous matrix, so $k_F = 0.0001 \ll k_m = 1$. Dealing with this new feature requires extensive modifications and reformulations of Section 2.2 since the node must be split along the fracture segment to form "double nodes", similar to the approach in Jiang and Yang (2018) and Segura and Carol (2004). Consequently, all of the 1D fracture element shape functions and their spatial derivatives are augmented to four columns. We omit the technical details and only display final results here because our focus is primarily on conductive fractures that satisfy the cross/transversal flow equilibrium (Jiang and Yang, 2018), such as hydraulic fractures in shale gas reservoirs. Nevertheless, we acknowledge that in geology, compaction bands can serve as flow barriers (Ip and Borja, 2023).

3.2. Anisotropic strip load problem

The second verification example examines the anisotropic strip load problem originally proposed by Zhao and Borja (2021). This example first validates the finite element implementation of the porous matrix, supplemented by a mesh sensitivity analysis to strengthen the reliability of the results. Finally, mirroring the approach in Section 3.1, we replace the original fluid with ideal gas to investigate how fluid type influences the strip load response.

The computational domain, including the finite element mesh and boundary/loading conditions, is shown in Fig. 7. All the material parameters can be obtained from the first column of `Table 1` in Zhao and Borja (2020), except that K_s is set to $+\infty$, which implies that the Biot tensor $\boldsymbol{b}$ downgrades to the scalar Biot coefficient of 1. We assume an over-consolidated (OC) rock with an initial preconsolidation pressure $p_{c0} = -2$ MPa (Zhao and Borja, 2020). The time step and traction boundary settings require further elaboration. Since ideal gas cannot have zero pressure, the reference pressure is set to 1 MPa in this example. Consequently, an in-situ uniform load of $F_a = -1$ MPa is applied to balance the initial fluid pressure. Subsequently, a uniform overburden of $F_b = -1$ MPa is applied in two steps, each with a large

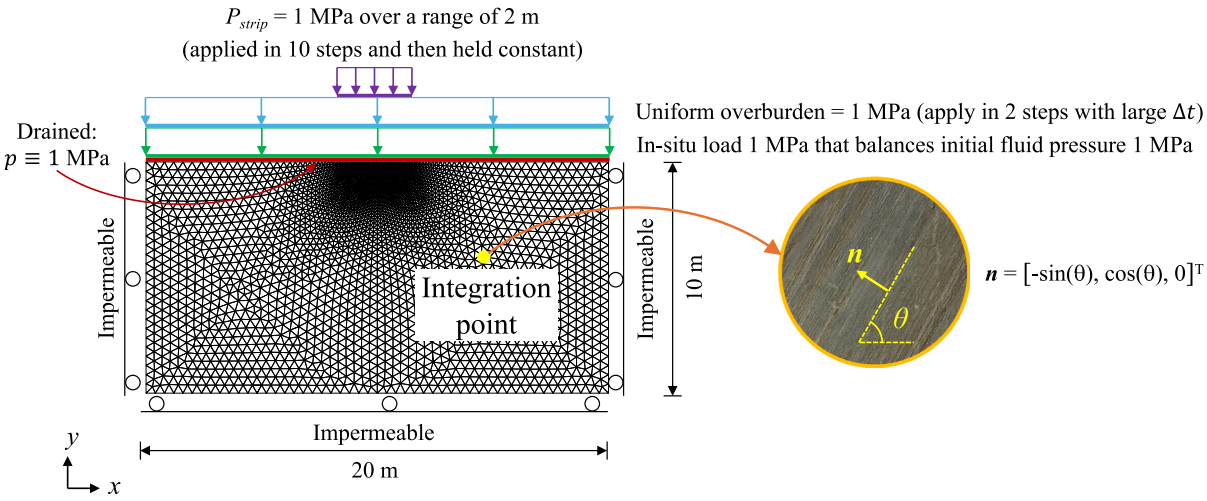


Fig. 7. Schematic diagram of the anisotropic strip load problem. Compared with Zhao and Borja (2020), the datum fluid pressure in this work is set to 1 MPa. Consequently, both the top drainage boundary and the initial fluid pressure are set to 1 MPa, which necessitates an in-situ uniform load (1 MPa) to balance the initial fluid pressure. All other loading sequences remain unchanged. Additionally, anisotropy is incorporated only within the flow and deformation constitutive updates at Gauss integration points and does not affect either the geometry or the mesh generation.

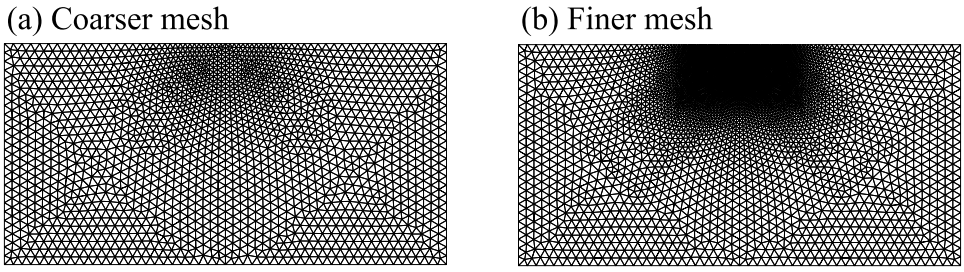


Fig. 8. Two other finite element meshes with different mesh densities are also adopted in this work for mesh sensitivity analysis.

time step of $\Delta t = 10000$ days. This approach results in permanent plastic deformation without generating any excess pressures. Next, the strip load $F_c = -1$ MPa is applied in 10 loading steps, each with a uniform time step of $\Delta t = 20$ minutes. Subsequently, the load is held constant for another 50 to 65 steps, during which the time step size is no longer uniform: at this stage, the initial time step length is set to 20 min, and for subsequent steps, the time increment is increased by a factor of 1.25 from the previous value, *i.e.*, $\Delta t_{n+1} = 1.25\,\Delta t_n$. For convenience in coding, the traction at the whole top boundary can be combined into the following compact form

$$F_y = F_a + F_b \times \min\left(1, \frac{t}{t_{\text{crit1}}}\right) + F_c \times 1_{x\in[9,11]} \times \min\left(1, \max\left(0, \frac{t - t_{\text{crit1}}}{t_{\text{crit2}} - t_{\text{crit1}}}\right)\right), \quad (33)$$

where $1_{x\in[9,11]}$ is an indicator function that equals 1 when x falls within the footing region (here, from $x = 9$ to $x = 11$), $t_{\text{crit1}} = 20000$ days represents the time instant when the overburden is fully applied, and $t_{\text{crit2}} = (20000$ days $+ 200$ min) denotes the time instant at which the strip load reaches its maximum.

We first verify our simulation results against those reported by Zhao and Borja (2020). In their study, the fluid is slightly compressible with a bulk modulus of 2 GPa. We reproduce the results on three meshes with different mesh densities, as shown in Fig. 8. The final distribution of the equivalent plastic strain, settlement curve, and excess pore pressure generation and dissipation curve are all combined into Fig. 9. Here the settlement curve is based on the temporal response of point (10, 10) m; while the excess pore pressure curve is based on the point (10, 9) m. In the figure, the "Time(day)" on the horizontal axis of the curve is actually measured starting from when the strip load is first applied, *i.e.*, $t - t_{\text{crit1}}$. The final equivalent plastic strain ε_p is calculated from the discrete sum of the incremental plastic strain tensor $\Delta\boldsymbol{\epsilon}^p$ as

$$\varepsilon_p = \sum_{\text{step}=1}^{N_{\text{step}}} \sqrt{\frac{2}{3}\,\Delta\boldsymbol{\epsilon}^p_{\text{step}} : \Delta\boldsymbol{\epsilon}^p_{\text{step}}}, \quad (34)$$

where N_{step} is the total number of time steps. That is to say, all the distributions of ε_p shown in this subsection are calculated from the last time step. Compared with the reference results, the settlement curve exhibits excellent agreement, and the excess pore-pressure curve is also largely consistent. Moreover, the equivalent plastic strain pattern remains consistent with the reference. The slight discrepancy in excess pore pressure at around 200 min can be attributed to the mesh density: a more diffuse strain localization would lead to a lower peak value of excess pore pressure. Therefore, the referenced work may have employed an even coarser mesh than Fig. 8(a).

Finally, in Fig. 10, we investigate the effect of fluid type on the strip load response (the fluid mobility is not changed). The results indicate that the induced excess gas pressure and subsequent settlement during the load-holding stage are negligible. The equivalent plastic strain distribution becomes slightly more diffuse compared to Fig. 9(b). These observations highlight that the stress path of the geomaterial may also depend on the type of filling fluid.

4. Model application

After verifying our model, we apply it to several synthetic cases. The first involves a triaxially confined, intact shale sample where the coupled process can be reduced to a nonlinear parabolic PDE for gas pressure under certain bedding plane orientations, allowing direct solution with MATLAB. The second case is a field scenario designed to

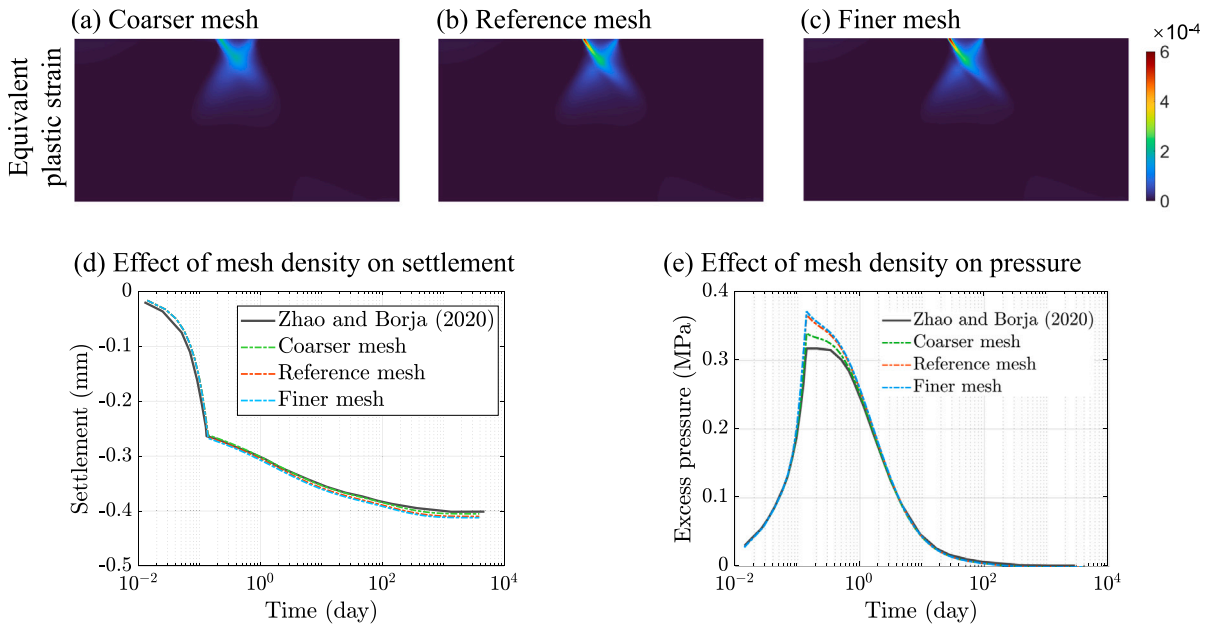


Fig. 9. Mesh sensitivity analysis results demonstrate a pronounced dependence on mesh density, particularly evident in the strain localization band width, maximum equivalent plastic strain, and maximum excess fluid pressure. Nevertheless, the settlement response exhibits minimal sensitivity to mesh density. Comparing our findings with those of Zhao and Borja (2020) suggests that the referenced work employs an even coarser mesh, as evidenced by the more diffused strain localization.

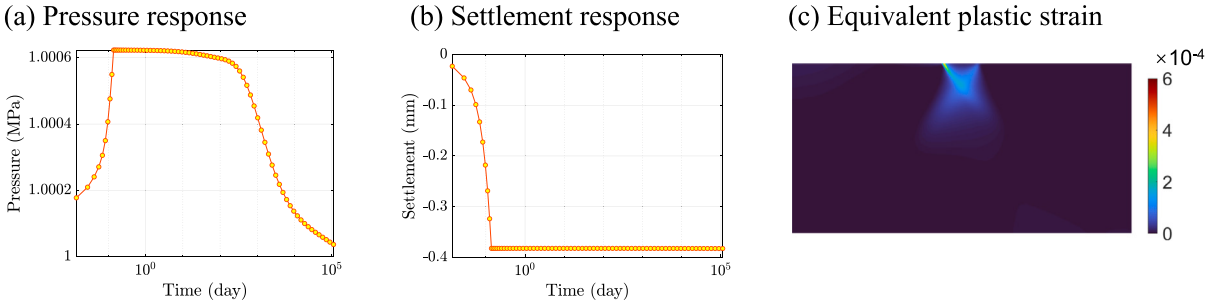


Fig. 10. Effect of the fluid type on strip load responses. The results indicate that the induced excess gas pressure and subsequent settlement during the load holding stage are negligible. The final equivalent plastic strain ϵ_p distribution becomes slightly more diffuse. These findings highlight the critical role of the fluid type in determining the response of porous media under external mechanical loading.

explore the effects of the bedding plane orientation and other factors on gas production from fractured shale reservoirs. In addition to horizontal and vertical bedding planes, we also examine an inclined bedding plane orientation. The third case concerns a more realistic 3D fractured shale reservoir.

4.1. One-dimensional gas production from triaxially confined intact shale

The purpose of the first example is to determine how geomechanics influences gas production. Here, we focus on intact shale to specifically examine gas flow within the porous matrix and minimize the number of adjustable parameters, thereby strengthening the credibility of the conclusions. In this example, we analyze the effect of geomechanics by comparing gas production across different combinations of the following factors: (1) bedding plane orientation, (2) non-Darcy flow, and (3) deformation mechanism, which can be further categorized as no deformation, elastic deformation, or elastoplastic deformation.

Fig. 11 shows the model schematic, which considers two specific bedding plane orientations: horizontal and vertical. These orientations ensure that the gas flow remains approximately one-dimensional, leading to a scalar permeability. Since the current non-Darcy flow equation applies only to scalar permeability (Zhang et al., 2019b), focusing on one-dimensional flow is essential for accurate modeling. Detailed material parameters are comprehensively recorded in Table 2. From the solid part, we could identify the shale to be quite soft (Yu and Sepehrnoori, 2014) and would have more pronounced deformation than the hard shale, which makes it perfect to study the geomechanics effect.

4.1.1. Simplified modeling approach

Instead of performing a fully three-dimensional simulation, we approximate the coupled governing equations by assuming that the total stress σ remains constant throughout the domain. This simplification reduces the 3D problem to a nonlinear parabolic PDE for gas pressure. Please note that this assumption is only an approximation and may not hold during the early production stage. However, it is commonly applied in modern reservoir simulations to account for solid deformation due to its simplicity and ease of implementation (Liu et al., 2018; Yan et al., 2020; Yu and Sepehrnoori, 2014). The essence in the simplification is that for a discrete list of gas pressure values, we can generate the corresponding list of Lagrangian porosity ϕ_L and apparent permeability k_a through the stress-driven point simulation. These lists form the look-up tables used in the PDE solver. For more details about the stress-driven point simulation, interested readers may refer to Box 1 of Zhang et al. (2024). Although Box 1 is designed for plane strain

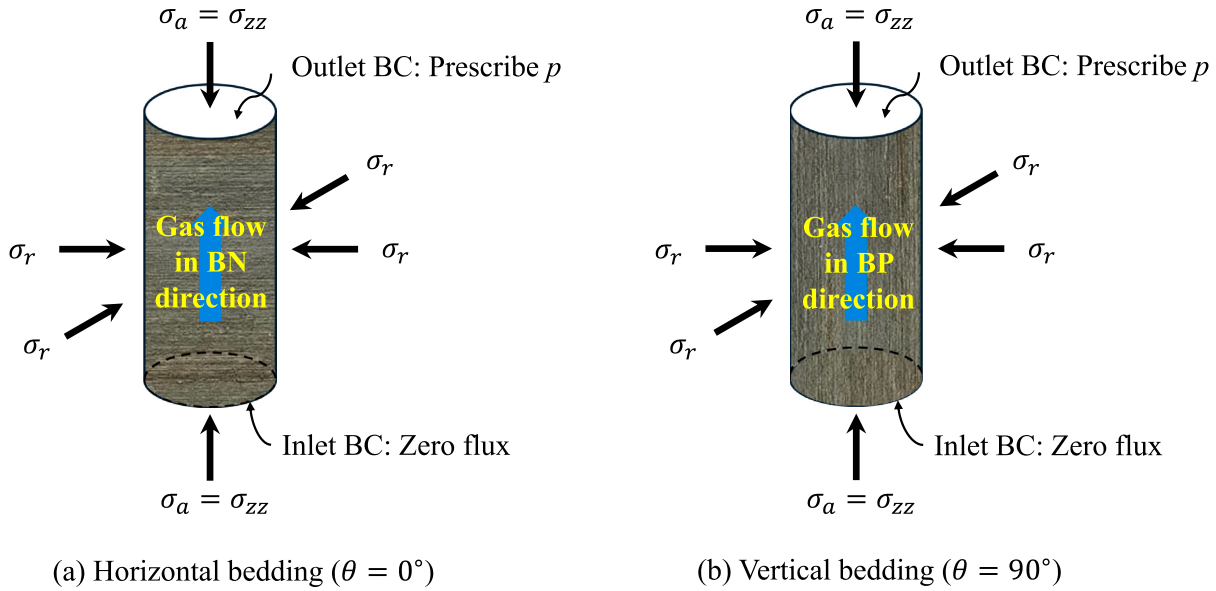


Fig. 11. Schematic diagram of one-dimensional gas production from triaxially confined intact shale. Two bedding plane orientations are considered. Abbreviations BC, BN, and BP represent boundary condition, bed-normal, and bed-parallel, respectively.

Table 2
Material parameters for one-dimensional gas production from triaxially confined intact shale. Abbreviations IC and BC stand for the initial conditions and the boundary conditions, respectively. The reference permeability k_0 is used in the normalization of apparent permeability k_a. In addition, the values of k_h, k_v, and $k_h/k_v = 20$ are consistent with the abstract summary in Ma et al. (2025a).

Parameter	Value	Unit
Solid:		
Young's moduli E_v, E_h	4.286, 7.3	GPa
Poisson's ratio ν_{hv}	0.15	1
Poisson's ratio ν_{hh} in the plane of isotropy	0.17	1
Shear modulus G_{vh}	2.17	GPa
Bedding normal $\boldsymbol{n}$	$\boldsymbol{e}_x$, $\boldsymbol{e}_z$	–
Plastic anisotropy parameter c_1^p, c_2^p, c_3^p	0.7, −0.36, 0.6	1
Slope of the critical state line M_{CSL}	1.07	1
Plastic compressibility parameter λ^p	0.05	1
Initial preconsolidation pressure p_{c0}	−20	MPa
Intrinsic bulk modulus of solid K_s	30	GPa
Gas flow:		
Initial matrix porosity ϕ_0	0.06	1
Reference permeability k_0	2×10^{-19}	m^2
Intrinsic permeability k_h	2×10^{-19}	m^2
Intrinsic permeability k_v	1×10^{-20}	m^2
Tortuosity τ	2	1
Model temperature T	80	°C
Gas viscosity μ_g	2×10^{-5}	Pa · s
Gas molar mass M_{gas}	16.042	g/mol
Gas molecule diameter d_m	4×10^{-10}	m
Langmuir pressure P_L	4	MPa
ϵ_L in sorption model	5×10^{-4}	1
Model IC and BC:		
Initial uniform gas pressure p_{ini}	30	MPa
Confining stress σ_r, axial stress σ_a	−35, −40	MPa
Outlet gas pressure	3	MPa

point simulation, it can be easily adapted for three-dimensional point simulation, where the overall process is actually more straightforward.

In this section, we introduce several dimensionless quantities and rewrite the PDE in a dimensionless form. We define the dimensionless pressure $\bar{u}$, dimensionless coordinate $\bar{z}$, and dimensionless time T_v as

$$\bar{u} = \frac{p}{p_{ini}}, \qquad \bar{z} = \frac{z}{L}, \qquad T_v = \frac{(k_0/\mu_g)\, p_{ini}\, t}{L^2}, \tag{35}$$

where p_{ini} is the initial gas pressure, L is the model length in the z direction, and k_0 is the reference permeability, which equals k_h in this example. Here, we assume μ_g is constant; otherwise, a reference value of μ_g would be used in the definition of T_v. The resulting dimensionless PDE is

$$\underbrace{c(\bar{u})}_{\partial(\phi_L p)/\partial p} \frac{\partial \bar{u}}{\partial T_v} = \frac{\partial}{\partial \bar{z}} \left(\bar{u}\, \bar{k}_a \frac{\partial \bar{u}}{\partial \bar{z}} \right), \tag{36}$$

where $c(\bar{u})$ represents a derivative that can be obtained via numerical differentiation, provided there are sufficient data points in the aforementioned look-up tables. The term $\bar{k}_a = \bar{k}_a(\bar{u}) = k_a(p)/k_0$ denotes the normalized permeability using k_0. In essence, all factors discussed in the first paragraph of Section 4.1 will only affect $c(\bar{u})$ and $\bar{k}_a(\bar{u})$ in Eq. (36). In addition to Eq. (36), the initial condition and boundary conditions are

$$\bar{u}(T_v = 0,\, \bar{z}) = 1\,, \quad \bar{u}(T_v, \bar{z}_{\text{out}}) = \frac{1}{10} + \frac{9}{10} \exp(-10^6 \times T_v)\,, \quad \bar{u}\, \bar{k}_a \left. \frac{\partial \bar{u}(T_v, \bar{z})}{\partial \bar{z}} \right|_{\bar{z}=\bar{z}_{\text{in}}} = 0\,, \tag{37}$$

where $\bar{z}_{\text{out}}$ indicates the outlet end, and $\bar{z}_{\text{in}}$ indicates the other end with zero flux. Note that 10^6 is an arbitrarily large penalty parameter used to prevent discontinuity of $\bar{u}$ with respect to T_v at the outlet end (Chen et al., 2025). Finally, upon solving the PDE, we obtain the dimensionless cumulative gas mass $\bar{m}_{gp}$ from the spatial distribution of $\bar{u}$ along $\bar{z}$ at any given time:

$$\bar{m}_{gp} = 1 - \int_0^1 \frac{\phi_L(\bar{u})}{\phi_0} \bar{u}\, \mathrm{d}\bar{z}\,, \tag{38}$$

where ϕ_0 is the initial porosity.

4.1.2. Evolution of apparent permeability and porosity

Fig. 12 plots the look-up tables of normalized permeability (using k_0) and Lagrangian porosity ϕ_L as a function of p. Compared to elastic deformation, plastic deformation induces additional pore compression, as illustrated in Fig. 12(c). This is also known as the localized pore compaction. Consequently, the apparent permeability after accounting for plastic deformation is lower than that without plastic deformation, as shown in Fig. 12(a) and (b). In both subfigures, the green curve always lies below the red and yellow curves, and the point where the green curve deviates from the red curve marks the yield point. Another

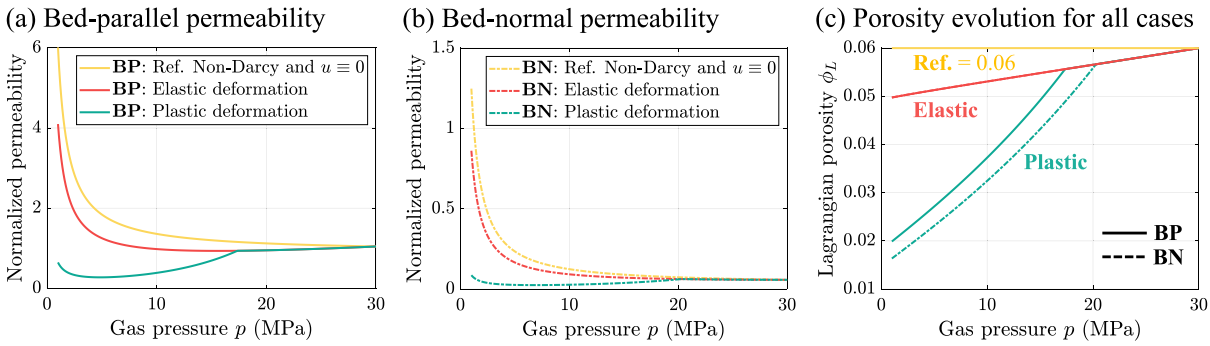


Fig. 12. Plot of BN and BP normalized permeabilities and Lagrangian porosity, all as functions of gas pressure. For normalized permeability, different deformation mechanisms play an obvious role. Lagrangian porosity ϕ_L shows differences between BN and BP flow directions only under plastic deformation.

interesting finding is that ϕ_L remains insensitive to the bedding plane orientation for purely elastic deformation.

In fact, as shown in Fig. 12(a) and (b), there is a subtle difference between the two, which can be highly important for subsequent gas production analysis. Because the intrinsic BN permeability k_v is only one-twentieth of the intrinsic BP permeability k_h, the non-Darcy effect is weaker for k_h and is more pronounced for k_v, as summarized below.

- Fig. 12(a): the green curve falls obviously below k_h, the left endpoint of the red curve is about 4 times k_h, and that of the yellow curve is roughly 6 times k_h.
- Fig. 12(b): the green curve remains close to k_v, whereas the left endpoint of the red curve is about 16 times k_v, and that of the yellow curve is about 25 times k_v. This indicates a more pronounced non-Darcy effect than Fig. 12(a).

These findings support the preceding discussion, and such differences will inevitably affect the subsequent gas production curves, as shown next.

4.1.3. Impact of geomechanics on gas production

We now plot $\bar{m}_{gp}$ against t_D for various scenarios in Fig. 13. In Fig. 13, we observe that the ordering of the green, red, and yellow curves differs between the BN and BP scenarios, which can be explained by the results in Fig. 12. For flow in the BP direction, the apparent permeability is notably smaller than k_h when plastic deformation is taken into account, as shown in Fig. 12(a), consequently, this resulting localized permeability reduction will serve as a barrier to fluid flow, affecting production, and that explains why the green solid curve lies below the other two solid curves for most of the time. By contrast, in the BN direction, the apparent permeability remains close to k_v when plastic deformation is considered, as shown in Fig. 12(b). Consequently, the barrier effect no longer exists or becomes very weak, causing the green dashed curve to lie above the yellow dashed curve in Fig. 13.

Then we compare the ultimate cumulative gas mass among different scenarios. Because the moduli (*e.g.*, E_v, E_h, and G_{vh}) are relatively small, the shale undergoes more pronounced compression, making the variations in total gas production induced by different deformation mechanisms more noticeable. As a result, as plastic deformation induces greater pore compression, its ultimate cumulative gas mass will inevitably exceed that of other scenarios (regardless of how various factors may reduce the permeability). In other words, the green production curve will eventually surpass the others and yield a higher cumulative output, as supported by Fig. 13. This trend is not reported in previous literature on geomechanical effects. Moreover, the angle of the bedding plane only affects the ultimate cumulative gas mass in the presence of plastic deformation, as only the solid and dashed green curves have different ultimate values.

In summary, we may draw the following conclusion: in existing literature, the decrease in production often reported after considering geomechanical effects can stem from two reasons. First, the time span investigated might be insufficiently long, and second, the permeability model used commonly undergoes a severe decrease in permeability when geomechanical effects are included. In other words, attributing decreased production solely to geomechanical effects without acknowledging the significant permeability reduction is scientifically unsubstantiated, and it may even be erroneous.

4.2. Field gas production from anisotropic fractured shale reservoir

As the second example in this study, a synthetic anisotropic fractured shale reservoir is presented in Fig. 14. Unlike the case in Fig. 11, this scenario incorporates additional complexities due to the presence of discrete fractures. Consequently, in addition to the bedding plane orientation, various fracture-related factors are also examined here. The simulation is performed under plane strain conditions, with gravity effects neglected. All the model parameters are provided in Table 3, from which we may observe that the shale property here is similar to that of Marcellus shale (Villamor Lora et al., 2016) and is harder than that in Table 2. Therefore, the localized shale matrix pore compaction would have negligible effects on the cumulative gas production, but there are other new emerging factors that affect gas production, as discussed in the following paragraphs.

It is essential to clarify the choice of boundary conditions in Fig. 14. In previous studies, only one quarter of the reservoir was modeled by imposing roller boundary conditions on the left and bottom boundaries, relying on a symmetry assumption. However, when anisotropy is introduced, this assumption is no longer valid. Consequently, we now treat the entire geometry depicted in Fig. 14 as the full reservoir. By fixing only the midpoint of the bottom boundary ($u_x = u_y = 0$ at that location) and applying $u_y = 0$ to the remaining points along the bottom boundary, we can more effectively capture and assess the impact of anisotropy.

4.2.1. Impact of the bedding plane orientation and finite element order on the reservoir gas pressure and deformation distributions

In Fig. 15, we illustrate the effect of anisotropy on the distributions of gas pressure, equivalent plastic strain calculated using Eq. (34), and horizontal displacement. Notable discrepancies are observed among different bedding plane orientations θ. Regarding the gas pressure field, when $\theta = \pi/2$, the low-pressure region expands predominantly in the vertical direction; by contrast, when $\theta = 0$, it expands primarily in the

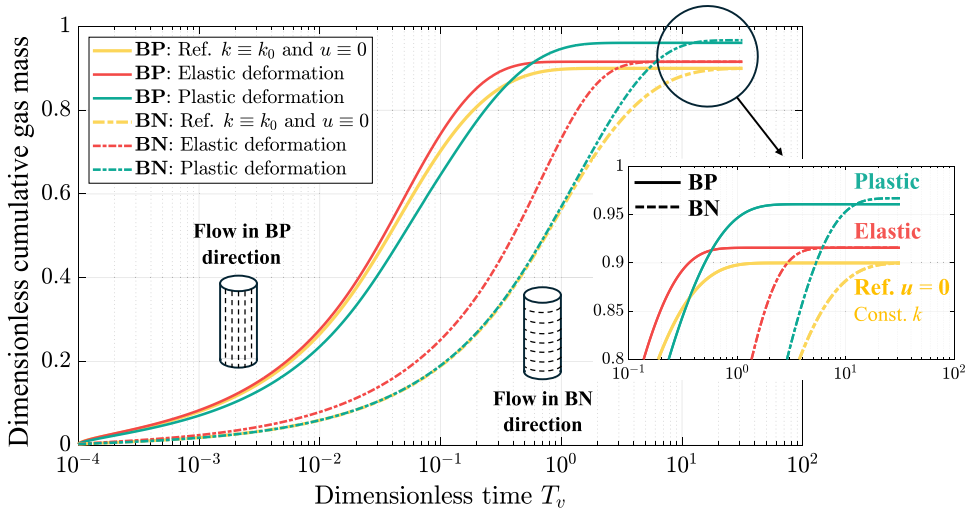


Fig. 13. A comprehensive gas production analysis over the entire production cycle, encompassing six different scenarios that vary in bedding plane orientation, permeability model, and deformation mechanism (elastic or plastic). The inset on the right offers a magnified view, providing a closer comparison of the ultimate gas production.

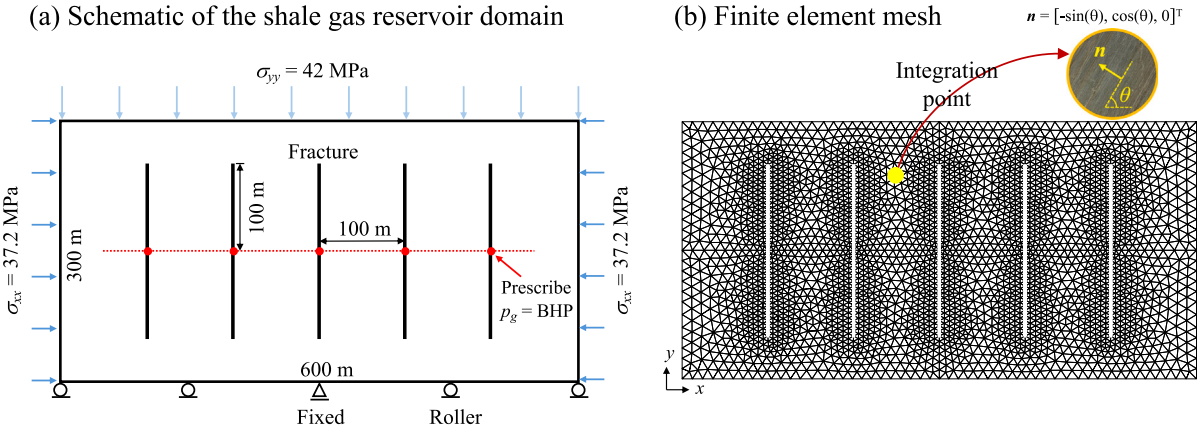


Fig. 14. Schematic diagram of field gas production from anisotropic fractured shale reservoir: geometry, boundary conditions, and the finite element mesh. The anisotropy is embedded in the same way as that for the anisotropic strip load problem. In the reference case, bottomhole pressure (BHP) is applied only at five intersection points where the vertical fracture meets the horizontal red dotted line. Later in the parametric analysis, we instead apply BHP along the entire horizontal red dotted line to examine its effect on gas production.

horizontal direction. When $\theta = \pi/4$, the pressure distribution is inclined from the lower-left corner to the upper-right corner. As discussed later in Section 4.2.2, these varying gas-pressure expansion patterns exert a substantial influence on cumulative gas production.

For the equivalent plastic strain, when $\theta = \pi/2$, plastic deformation is concentrated around the fracture, whereas for $\theta = 0$, it is not concentrated around the fracture. In Fig. 16, we present the final equivalent plastic strain distribution, where the differences become even more pronounced; specifically, for $\theta = 0$, the matrix region between two adjacent fractures experiences greater plastic deformation. This result is analogous to different shear band orientations (parallel or perpendicular to the bedding plane) during the compression of transversely isotropic shale specimens (Zhao et al., 2018; Borja et al., 2020).

Fig. 17 investigates the effect of the finite element order, which can be interpreted as a p-refinement study. The findings indicate that only the equivalent plastic strain becomes more pronounced and localized with higher-order elements, whereas the other fields remain effectively unchanged, thereby reinforcing the reliability of our results.

Regarding horizontal displacement, an inclined distribution is observed when $\theta = \pi/4$, as expected. Nevertheless, an unanticipated phenomenon emerges: the u_x distribution in column (c) is skewed, whereas it remains relatively symmetrical in column (a). To elucidate this, Fig. 18 compares vertical compression between two cases selected for their similar cumulative gas mass production (m_{gp}). Specifically, in Fig. 18, the left column (a) has $m_{gp} = 0.576 \times 10^6$ kg, and the right column (b) has $m_{gp} = 0.57 \times 10^6$ kg, both of which are sufficiently close. The results show that, for the horizontal bedding configuration, overall vertical compression is more pronounced than in the reference vertical bedding configuration. Because the non-uniformity of u_y along the line B–B' is greater than along A–A' or C–C', u_x becomes skewed and asymmetric about the horizontal centerline $y = 150$ m in column (b), a consequence of the Poisson effect.

4.2.2. *Reservoir cumulative gas production analysis*

In Fig. 19, we conduct the gas production analysis considering a variety of factors. For subfigure (a), in the reference case ($\theta = \pi/2$), when plotted on a logarithmic time scale, two plateau-like stages in the production curve become evident. The first plateau occurs around 10 days and exhibits a tiny amplitude, as it reflects gas production from the narrow banded region containing fractures. During this stage, the pressure in the rest of the shale matrix remains nearly unchanged, and additional time is needed for the matrix pressure to decline. Consequently, from 10 days to 100 days, the production curve remains almost flat. The second plateau indicates the completion of shale gas

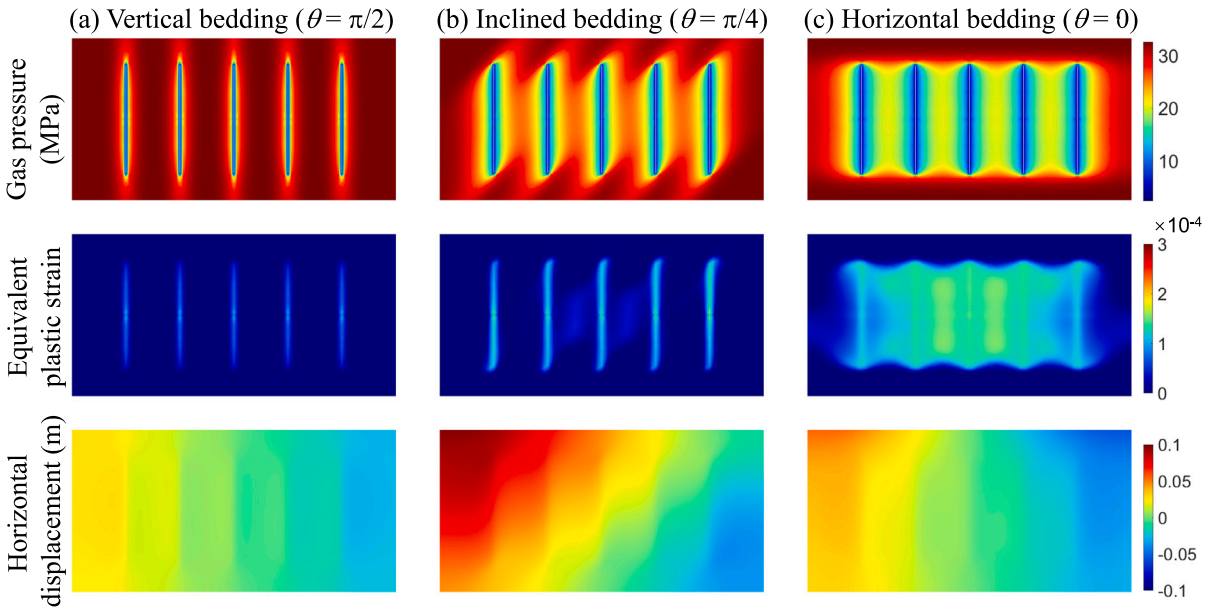


Fig. 15. Effect of anisotropy on gas pressure, equivalent plastic strain, and horizontal displacement distributions at t = 1391 days (77th step). Equal-order linear interpolation between displacement field and gas pressure field is adopted.

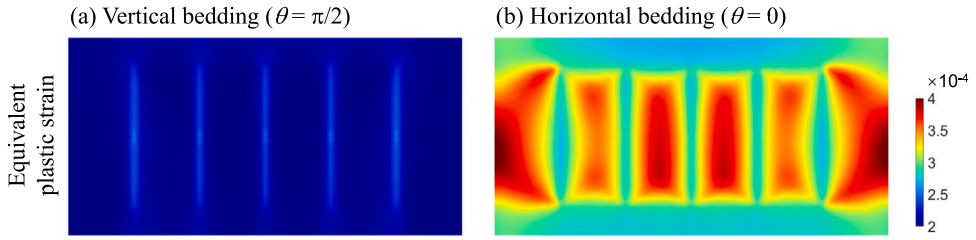


Fig. 16. Effect of anisotropy on the final equivalent plastic strain distribution (130th step). Two different bedding plane orientations lead to strikingly different plastic deformation patterns. The patterns are also completely different from the case of pure solid deformation, *i.e.*, increase σ_{xx} and σ_{yy} until the target value while keeping pressure field unchanged at the BHP value.

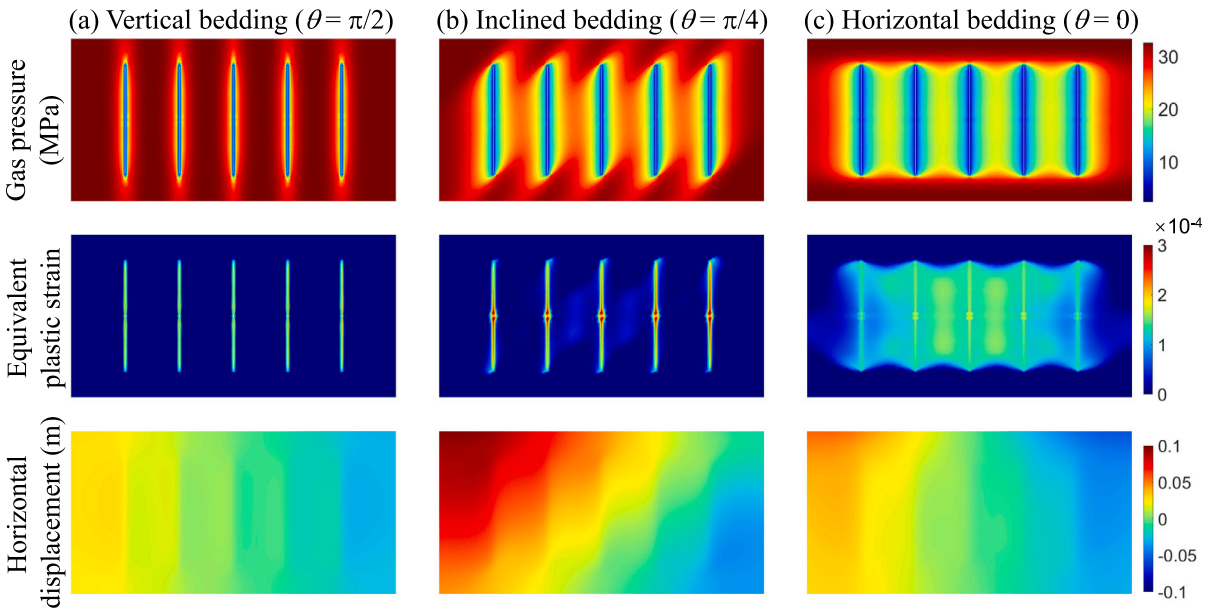


Fig. 17. The p-refinement results at the same time instant as the previous Fig. 15. The displacement is interpolated using the six-node triangular element. As expected, only the equivalent plastic strain becomes more pronounced and localized.

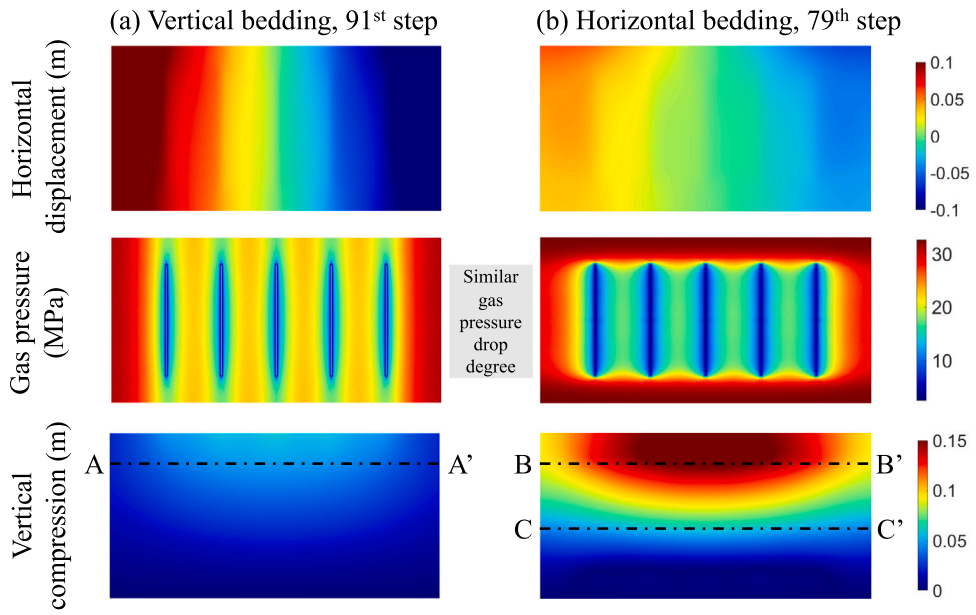


Fig. 18. Explanation of the much more skewed u_x contour in the horizontal bedding case than that in the vertical bedding case. To ensure a fair comparison, both cases were analyzed at similar gas pressure drop degrees, with the vertical bedding case requiring more time to achieve this condition.

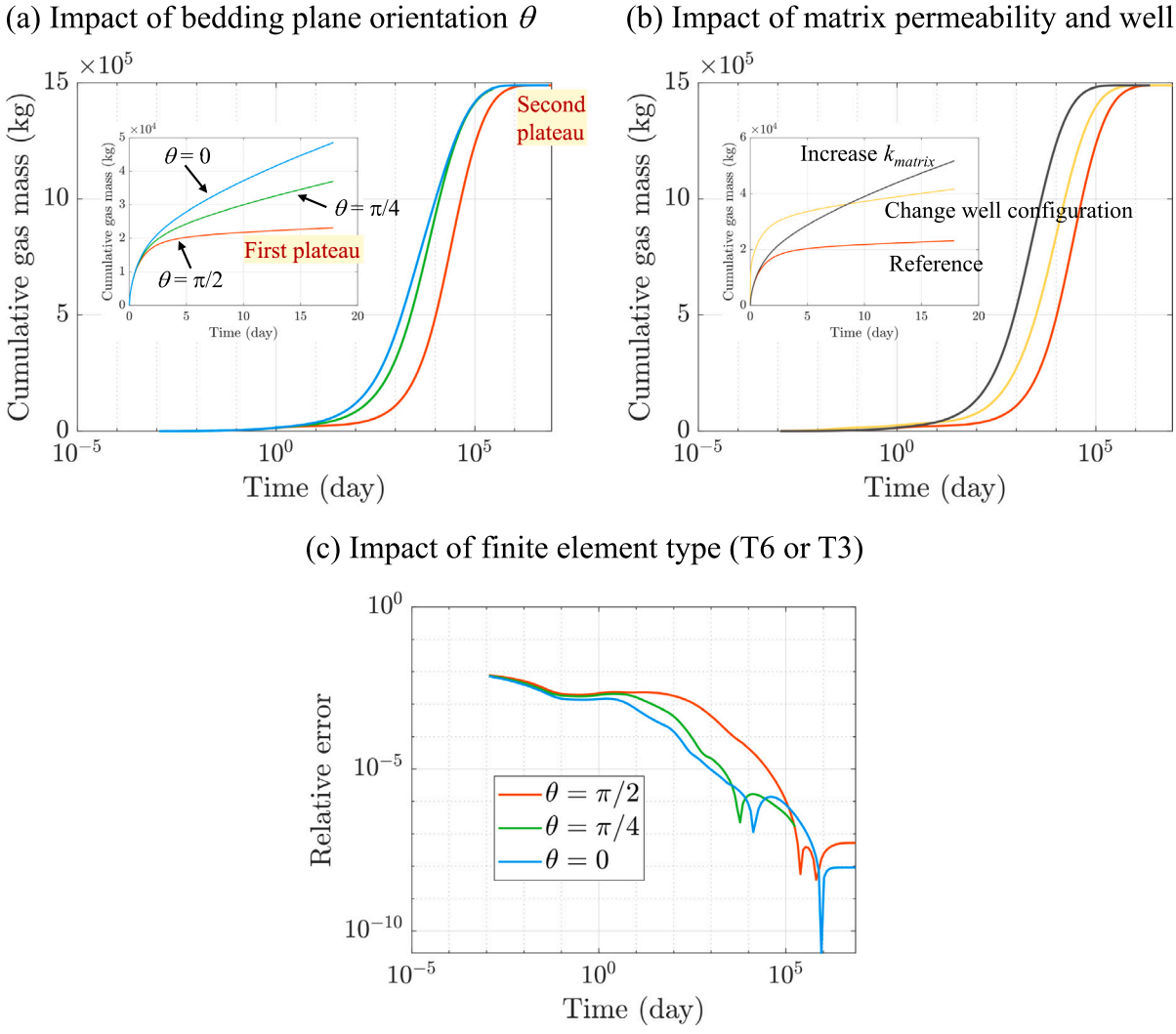


Fig. 19. A comprehensive gas production analysis considering a variety of factors including the bedding plane orientation θ, matrix permeability, well configuration, and element type.

Table 3
Model parameters of a synthetic anisotropic fractured shale reservoir. For bedding plane orientation θ, the bedding normal is constructed as $\boldsymbol{n} = \left[-\sin\theta, \cos\theta, 0\right]^{\mathsf{T}}$, and the full-tensor matrix permeability is therefore $\boldsymbol{k} = k_h(\mathbf{1} - \boldsymbol{n}\otimes\boldsymbol{n}) + k_v\,\boldsymbol{n}\otimes\boldsymbol{n}$.

Parameter	Value	Unit
Solid:		
Young's moduli E_v, E_h	19, 31	GPa
Poisson's ratio ν_{vh}	0.24	1
Poisson's ratio ν_{hh} in the plane of isotropy	0.3	1
Shear modulus G_{vh}	13	GPa
Bedding plane orientation θ	0, $\pi/4$, $\pi/2$	rad
Plastic anisotropy parameter c_1^p, c_2^p, c_3^p	0.72, −0.36, 0.56	1
Slope of the critical state line M_{CSL}	1.07	1
Plastic compressibility parameter λ^p	0.0005	1
Initial preconsolidation pressure p_{c0}	−20	MPa
Biot coefficient	1	1
Gas flow:		
Porosity ϕ	0.05	1
Permeability k_h	5×10^{-19}	m^2
Permeability k_v	5×10^{-20}	m^2
Gas viscosity μ_g	2×10^{-5}	Pa · s
Model temperature T	80	°C
Gas molar mass M_{gas}	16.042	g/mol
Fracture flow:		
Fracture permeability k_F	8.33×10^{-12}	m^2
Fracture aperture w	0.001	m
Fracture porosity ϕ_F	0.75	1
Other information:		
Initial uniform effective stress σ'_{xx}, σ'_{yy}, σ'_{zz}	−4.6, −9.4, −4.6	MPa
Initial uniform gas pressure P_0	32.6	MPa
Ultimate gas pressure drop ΔP	30.2	MPa
Prescribed bottomhole pressure (BHP)	$P_0 - \Delta P\left(1 - e^{-t/t_d}\right)$	MPa
Parameter t_d that controls pressure drop	2500	s
Initial time step Δt_1	100	s
Subsequent time step Δt_{n+1}	1.175 Δt_n	s
Total simulation steps	130	1

production across the entire domain, demonstrating a dramatically different time scale compared to the first plateau.

However, under the condition $\theta = 0$, the first plateau becomes nearly imperceptible. This results from the high horizontal permeability, which causes the pressure in the surrounding shale matrix to decrease simultaneously with gas production in the narrow banded region containing fractures. Hence, there is no delay before the matrix contributes to gas production, leading to a steeper production curve from 10 days to 100 days than is observed in the reference case.

In subfigure (b), we investigate the effect of the bottom-hole pressure (BHP) boundary and the magnitude of matrix permeability $\boldsymbol{k}$ on cumulative gas production, by imposing BHP along the entire horizontal red dotted line in Fig. 14 and increasing the permeability tensor $\boldsymbol{k}$ by a factor of ten, respectively. The results indicate that modifying the well configuration yields higher early-stage gas production compared to increasing matrix permeability; however, at later stages, improving matrix permeability has a greater impact on overall gas production. Lastly in subfigure (c), we calculate the relative error in cumulative gas mass induced by the p-refinement study; the small magnitude of this error further supports the reliability of our findings.

As an aside, while not shown here, we also investigate the effect of BHP under the case of horizontal bedding $\theta = 0$. It turns out the enhancement in gas production is much weaker than the reference case of vertical bedding. This could be well justified. In the horizontal bedding case, the gas preferential flow direction is horizontal, therefore, even though the BHP is prescribed along the entire horizontal red dotted line, vertical conduction remains limited due to the relatively low permeability in the vertical direction. In contrast, in the reference case of vertical bedding with the new pressure boundary, vertical conduction is considerably enhanced, resulting in a more pronounced increase in gas production.

Table 4
Final cumulative gas mass at the 130th simulation step for various scenarios as described in Fig. 19. Although almost negligible, the bedding plane orientation is the only factor that affects the final cumulative gas mass in our setting, which is consistent with our intuition and thus from another perspective, verifies the reasonableness of the numerical simulator. This conclusion is also consistent with the trend in Fig. 13.

Case	Subcase	Cumulative gas mass (10^6 kg)
Vertical bedding	Reference	1.488961
	T6 FEM	1.488961
	Increase $\boldsymbol{k}_{\text{matrix}}$	1.488961
	Change well configuration	1.488962
Horizontal bedding	Basic T3 FEM	1.489054
	T6 FEM	1.489054
	Change well configuration	1.489055

Table 4 presents the final cumulative gas mass at the last time step for the various scenarios described in Fig. 19. Because the moduli in Table 3 are substantially larger than those in Table 2, variations in total gas production due to bedding plane orientation are not readily discernible from the figure and are, therefore, primarily conveyed through the tabulated data. Nevertheless, the total gas production does vary with changes in bedding plane orientation, even under these large-modulus conditions.

4.3. More realistic practical example

Despite the above extensive discussions, the results are still theoretical in nature. Therefore, in this final subsection, we give a more practical three-dimensional field example to validate the previous conclusions on geomechanical effects. In that way, this study would have more practical applicability.

Fig. 20 depicts the model schematic and the corresponding finite element mesh. Compared with Fig. 14, the fracture patterns are more realistic, which contain both hydraulic fractures and induced branching fractures. The hydraulic fractures could be inclined as well. The production well is located at the center of the hydraulic fracture. The uniform constant load (30 MPa in compression) is applied on the top, right, and back boundaries, and the remaining bottom, left, and front boundaries are supported by rollers. The model parameters are inherited from Table 3 with the following changes: moduli values were reduced to 1/10 of their original magnitudes, initial porosity $\phi = 0.1$, initial reservoir gas pressure $P_0 = 30$ MPa, initial effective stress is zero, bottomhole pressure BHP = 5 MPa, axis of cross anisotropy of the bedding plane $\boldsymbol{n} = \boldsymbol{e}_z$. The induced branching fractures have the porosity, aperture, and constant permeability of 0.4, 0.25 mm, and 6.25×10^{-14} m^2, respectively. The stress-dependent normalized hydraulic fracture conductivity is shown in Fig. 21. The final gas production curve with and without geomechanical effects is given in Fig. 22. For the fully coupled case (with geomechanics), the typical distributions of various field quantities are summarized in Fig. 23.

To conclude, with the geomechanical effects, the hydraulic fracture conductivity loss given in Fig. 21 will serve as a barrier to fluid flow and affect production **during the practical production period**; while when $t \to \infty$, because the matrix deformation (geomechanical effects) induces greater pore compression, its ultimate cumulative gas mass will inevitably exceed that of the scenario with no deformation (regardless of how various factors may reduce the fracture permeability), and that explains the red line finally approaches and surpasses the blue line. This result aligns with the observations in Fig. 13, enhancing the practical applicability of our work.

Last but not least, since the main contribution of this work lies in the incorporation of a compressible gas, so we are also curious if the compressible gas is substituted by a slightly compressible or incompressible fluid such as water, whether the aforementioned trend in Fig. 22 would still hold. A short answer to this question is "No", and more details are given in Appendix B.

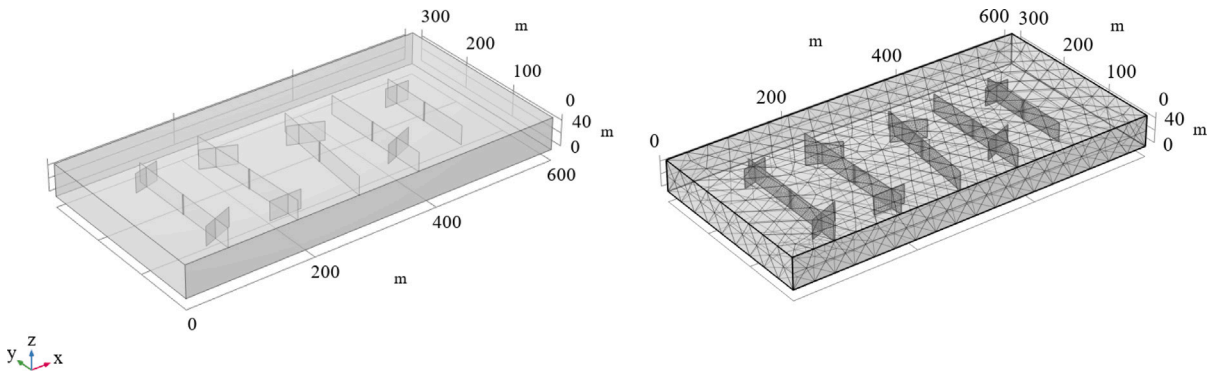


Fig. 20. Schematic of a more practical three-dimensional field example. The fracture patterns are more realistic, which contain both hydraulic fractures and induced branching fractures.

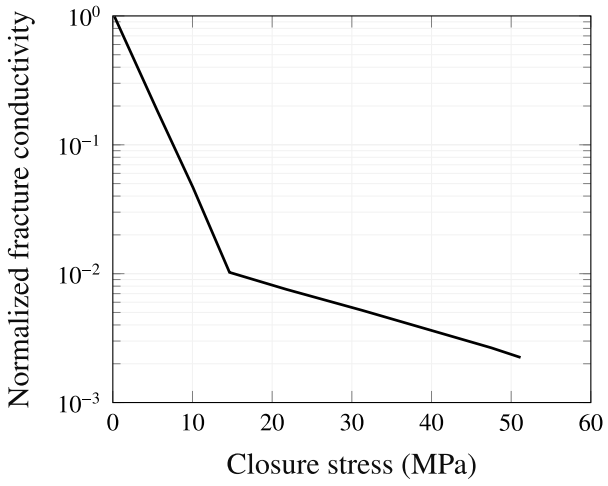


Fig. 21. Normalized hydraulic fracture conductivity under different "effective" closure stress. This pronounced loss of conductivity can be attributed to formation damage (Xu et al., 2016) resulting from proppant embedment, residual fracturing fluid, and proppant fines migration.

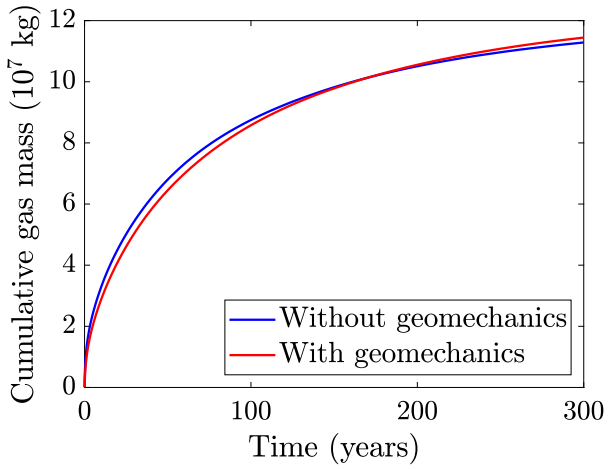


Fig. 22. Cumulative gas production from the practical three-dimensional field reservoir, which discusses the so-called geomechanical effects on gas production.

5. Closure

Shale commonly exhibits anisotropy, and the coupled processes of gas flow and solid deformation are pivotal for optimizing shale gas exploration and development. This paper provides a numerical framework for simulating gas flow and solid deformation in unconventional anisotropic shale. A rigorous mixed $\boldsymbol{u}/p$ formulation coupled with a one-dimensional finite element discretization of the fracture gas flow equation is adopted as the main numerical approach. The novel contributions are the incorporation of a compressible gas in the pores and the corresponding numerical ramifications, which guarantee the asymptotic quadratic convergence in the finite element solutions. Numerical verifications on discrete fracture network and anisotropic strip load problems demonstrate satisfactory performance. The impact of the filling fluid type is also revealed. Finally, model application offers a more "unbiased" perspective on the geomechanical effects influencing cumulative gas production, together with new trend in the plastic strain distribution during gas production associated with different bedding plane orientations.

Therefore, in general, it is unwarranted to assume that incorporating geomechanics invariably reduces gas production. Instead, one must investigate factors such as the time span, permeability decay, additional porosity reduction, and initial gas pressure overshoot to rigorously identify the specific mechanisms driving production changes.

CRediT authorship contribution statement

Qi Zhang: Writing – original draft, Methodology, Formal analysis, Writing – review & editing, Validation, Investigation. **Xia Yan:** Writing – review & editing, Methodology, Project administration. **Xinyu Wang:** Visualization, Methodology, Writing – review & editing, Software, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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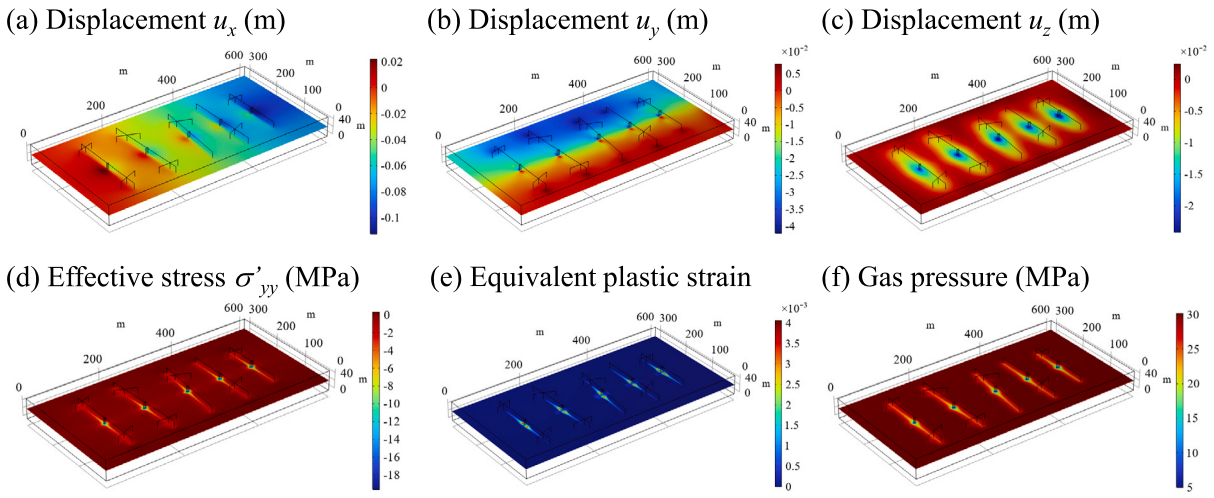


Fig. 23. Distributions of displacement in three directions, effective stress in the y-direction, equivalent plastic strain, and gas pressure are shown for the mid-depth cross section ($z = 25$ m) at an early production time of $t = 77.8$ days.

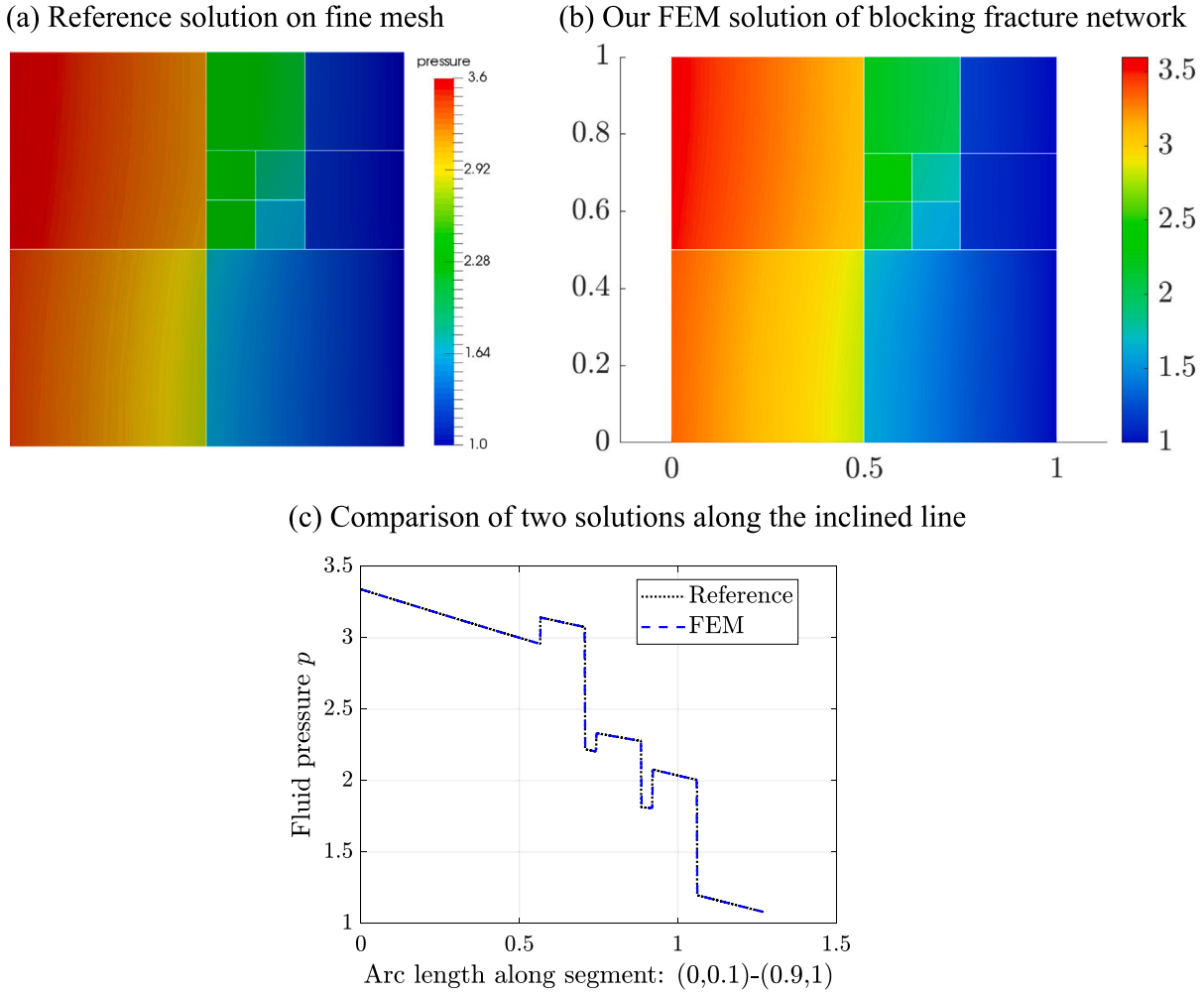


Fig. A.24. Another verification for the blocking fracture network. In this case, the transversal fracture flow induced by pressure discontinuity must be taken into consideration. This enhanced functionality is currently under development. For this benchmark problem, as depicted here, our results have successfully matched the reference solution. Quantitative errors and other information are reported in Table 1.

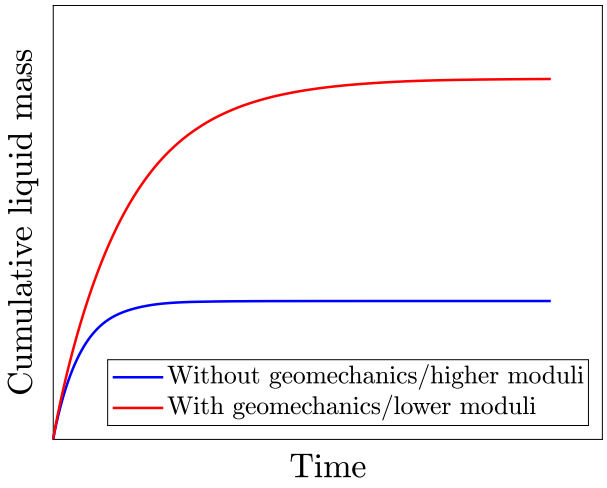


Fig. B.25. Geomechanical effects on the production of slightly compressible fluid.

Appendix A. Additional case of the blocking fracture

In this appendix, we show the verification results for the blocking fracture network in Fig. A.24 and Table 1. The main challenge associated with the node splitting technique is the allocation of degrees of freedom for newly added nodes, especially for the intersection node. Currently, our in-house simulator could handle the case of cross intersection and T-shaped intersection of two straight fractures (see Fig. 3), while other more complicated scenarios are left for future work.

Appendix B. Geomechanical effects on the production of slightly compressible fluid

For slightly compressible fluid, the compression of pore spaces plays the most important role in the cumulative fluid production, rather than the fluid's own compressibility. Therefore, although incorporating geomechanical effects or reducing the moduli can lead to greater pore compaction, thereby decreasing permeability to some extent, the negative impact of this permeability reduction is far outweighed by the production increase resulting from additional pore compaction. As a result, the net effect of considering geomechanical effects is still an increase in total production. This behavior is summarized in Fig. B.25, where it can also be observed that smaller moduli prolong the gas production period, thus delaying the time required to reach the horizontal stage of the production curve. To further illustrate this principle, consider an extreme scenario with an incompressible fluid and zero compression of pore spaces: under these conditions, the system would immediately reach steady-state and no production could be measured, *i.e.*, the total production would be zero. In addition, the trend in Fig. B.25 is also supported by `Figure 20` in Zhang (2020) and `Figure 21` in Yan et al. (2018b), whose subject of study is slightly compressible fluid with compressibility on the order of 10^{-9} Pa^{-1}.

Data availability

Data and codes are publicly available at https://gitee.com/qzhang94/fem-and-pde-tool-for-fractured-shale.

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